

Supporting Information for:  
Convolutional Neural Networks and Volcano  
Plots: Screening and Prediction of  
Two-Dimensional Single-Atom Catalysts for CO<sub>2</sub>  
Reduction Reactions

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# 1 Additional details on DFT calculations

## 1.1 Catalyst models and their thermodynamical stability

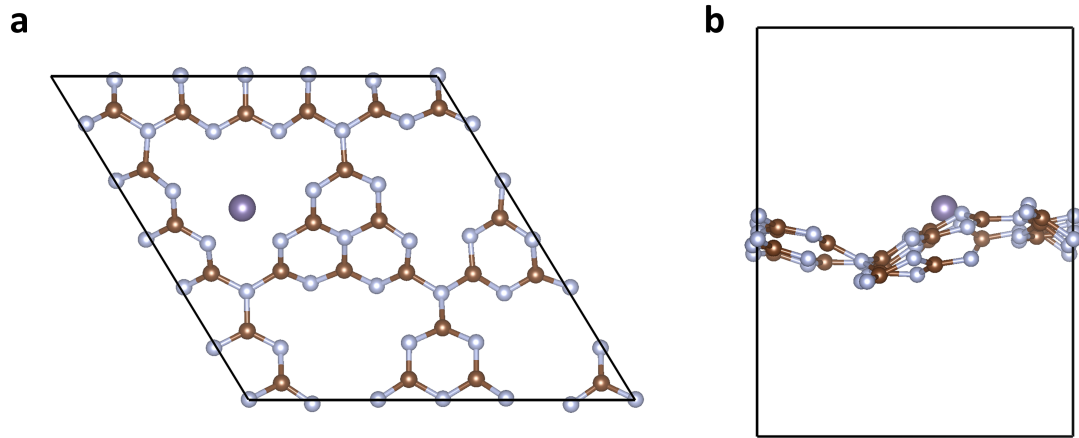
Representative structures and thermodynamical stability for each substrate are depicted in Figure 2 through Figure 13. For the adsorption process, the initial state is defined by positioning the optimized adsorbates at a distance of 6.5 Å from the single-metal atoms supported on the substrate, oriented along the z-axis. The thermodynamical stability is analyzed by calculate the energy change of the single metal atom deposition process, where a more negative energy change indicates more stable catalyst.

|    |    |    |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|----|----|----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 1  |    |    |     |     |     |     |     |     |     |     |     |     |     |     |     |     | 2   |
| H  |    |    |     |     |     |     |     |     |     |     |     |     |     |     |     |     | He  |
| 3  | 4  |    |     |     |     |     |     |     |     |     |     | 5   | 6   | 7   | 8   | 9   | 10  |
| Li | Be |    |     |     |     |     |     |     |     |     |     | B   | C   | N   | O   | F   | Ne  |
| 11 | 12 |    |     |     |     |     |     |     |     |     |     | 13  | 14  | 15  | 16  | 17  | 18  |
| Na | Mg |    |     |     |     |     |     |     |     |     |     | Al  | Si  | P   | S   | Cl  | Ar  |
| 19 | 20 | 21 | 22  | 23  | 24  | 25  | 26  | 27  | 28  | 29  | 30  | 31  | 32  | 33  | 34  | 35  | 36  |
| K  | Ca | Sc | Ti  | V   | Cr  | Mn  | Fe  | Co  | Ni  | Cu  | Zn  | Ga  | Ge  | As  | Se  | Br  | Kr  |
| 37 | 38 | 39 | 40  | 41  | 42  | 43  | 44  | 45  | 46  | 47  | 48  | 49  | 50  | 51  | 52  | 53  | 54  |
| Rb | Sr | Y  | Zr  | Nb  | Mo  | Tc  | Ru  | Rh  | Pd  | Ag  | Cd  | In  | Sn  | Sb  | Te  | I   | Xe  |
| 55 | 56 | La | 72  | 73  | 74  | 75  | 76  | 77  | 78  | 79  | 80  | 81  | 82  | 83  | 84  | 85  | 86  |
| Cs | Ba |    | Hf  | Ta  | W   | Re  | Os  | Ir  | Pt  | Au  | Hg  | Tl  | Pb  | Bi  | Po  | At  | Rn  |
| 87 | 88 | Ac | 104 | 105 | 106 | 107 | 108 | 109 | 110 | 111 | 112 | 113 | 114 | 115 | 116 | 117 | 118 |
| Fr | Ra |    | Rf  | Db  | Sg  | Bh  | Hs  | Mt  | Ds  | Rg  | Cn  | Nh  | Fl  | Mc  | Lv  | Ts  | Og  |

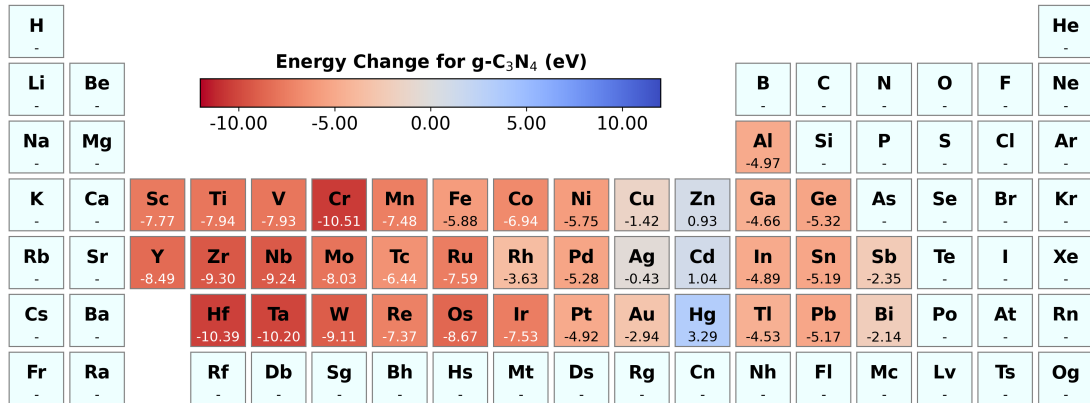
**Figure 1: Periodic table of investigated metals.** Periodic table highlighting investigated elements: transition metals in blue, metalloids in purple, and other metals in orange.

## 1.2 CO<sub>2</sub>RR pathways

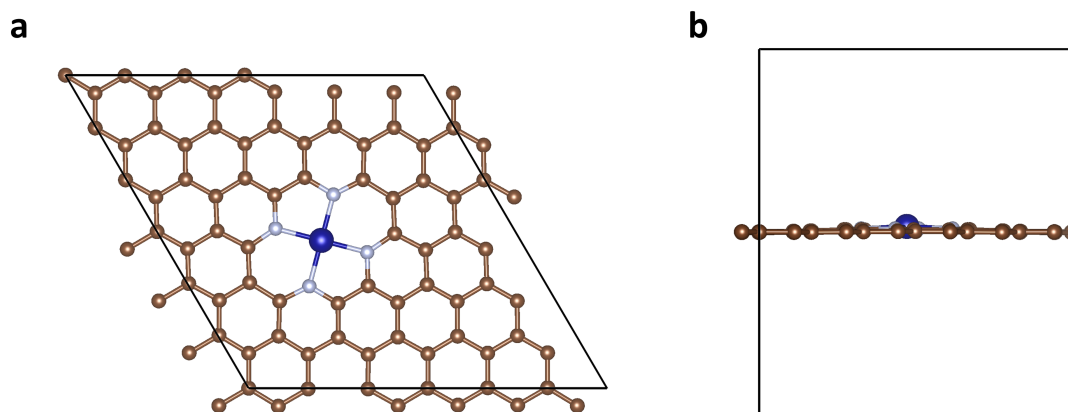
In this study, we explored three reaction pathways that have been previously documented in the literature [1, 2, 3], as depicted in Figure 14. The adsorption configuration of the \*CHO intermediate is pivotal in each of these pathways for determining the reaction mechanism in the CO<sub>2</sub>RR process. We calculated the energies associated with the \*CHO intermediate for catalysts supported on g-C<sub>3</sub>N<sub>4</sub>, nitrogen-doped graphene, and dual-vacancy graphene-under each mechanism. These calculations are summarized in section 1.2. Our results indicate that Pathway 1 is the most energetically favorable for most catalysts we investi-



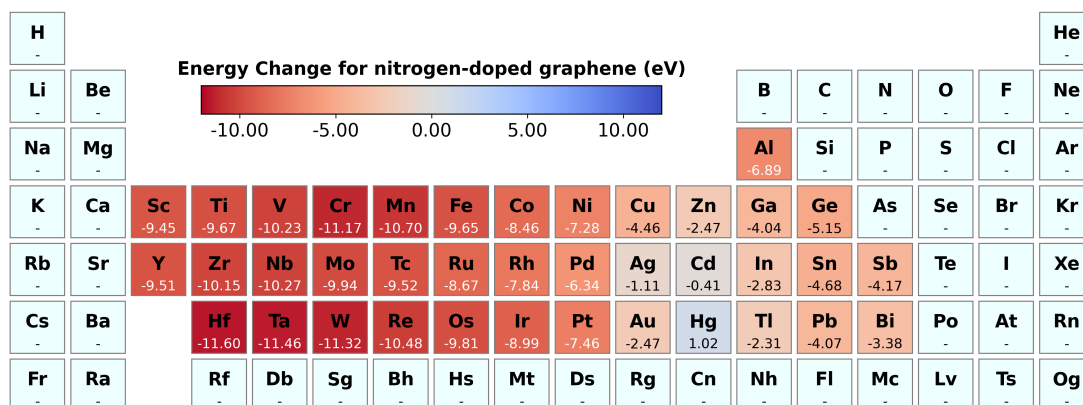
**Figure 2: Structure of single germanium atom supported on graphitic nitride.** (a) View along the z-axis and (b) view along the x-axis of the Ge@g-C<sub>3</sub>N<sub>4</sub> structure. In the illustrations, silver spheres represent N atoms, brown spheres signify C atoms, and purple sphere indicates Ge atom.



**Figure 3: Stability of single metal atoms supported on graphitic nitride.**

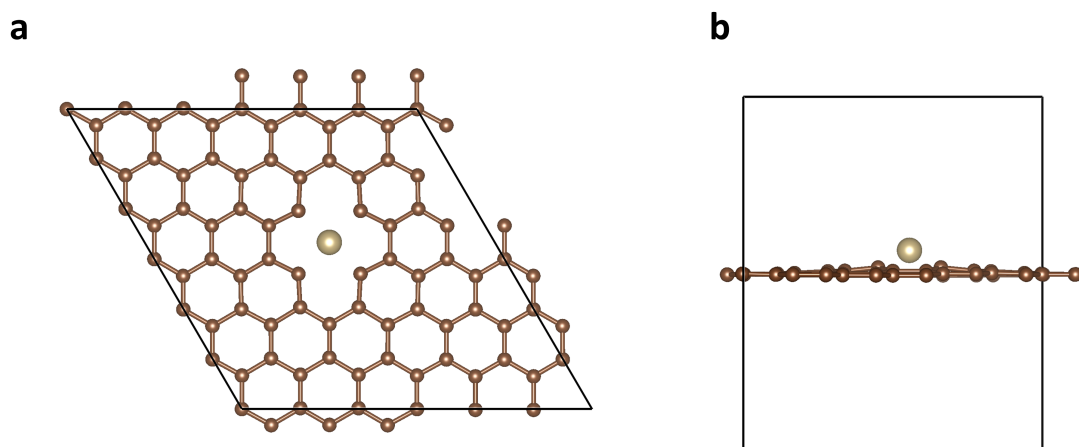


**Figure 4: Structure of single chromium atom supported on nitrogen-doped graphene.** (a) View along the z-axis and (b) view along the x-axis of the Cr@nitrogen-doped-graphene structure. In the illustrations, silver spheres represent N atoms, brown spheres signify C atoms, and indigo sphere indicates Cr atom.

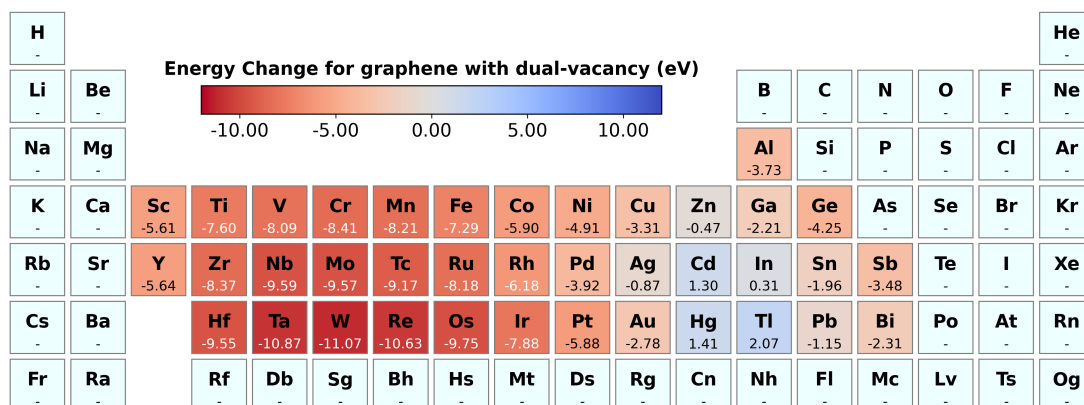


**Figure 5: Stability of single metal atoms supported on graphitic nitride.**

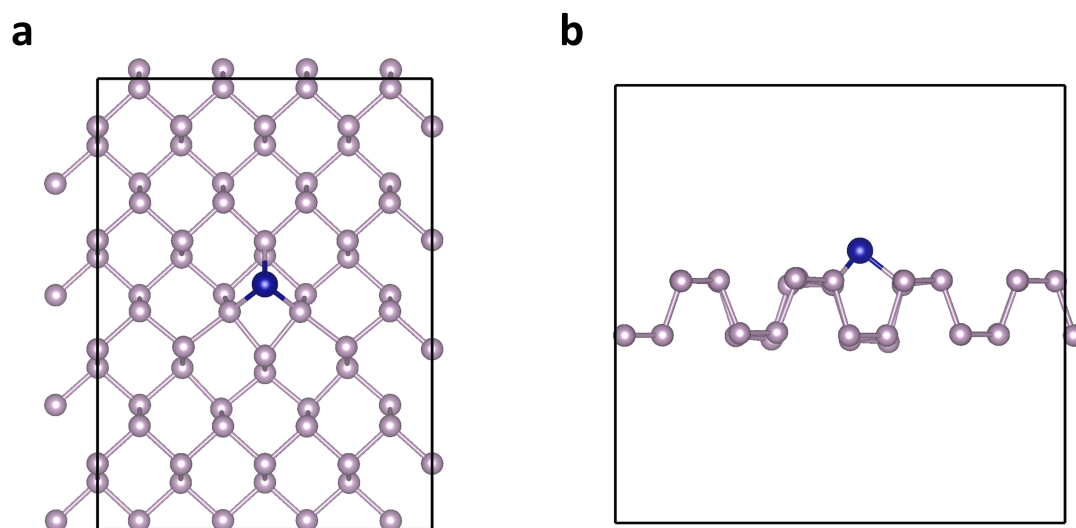




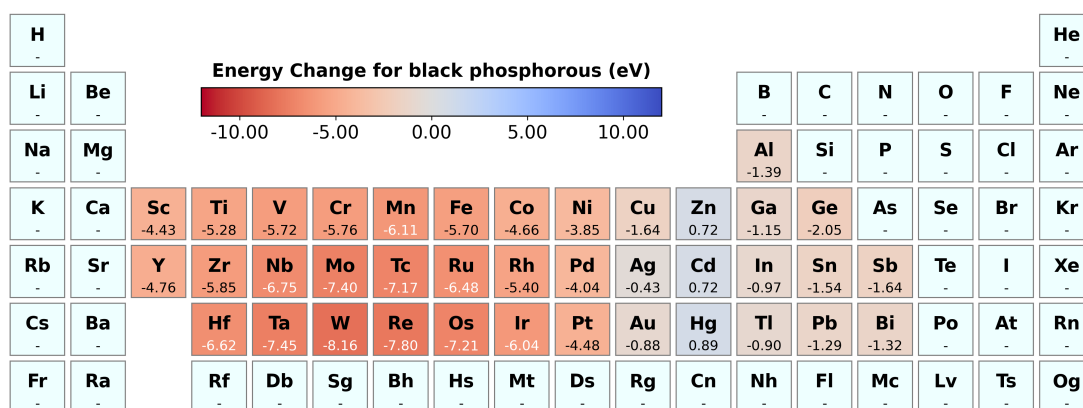
**Figure 6: Structure of single osmium atom supported on graphene with dual-vacancy.** (a) View along the z-axis and (b) view along the x-axis of the Os@graphene-with-dual-vacancy structure. In the illustrations, brown spheres signify C atoms, and light yellow sphere indicates Os atom.



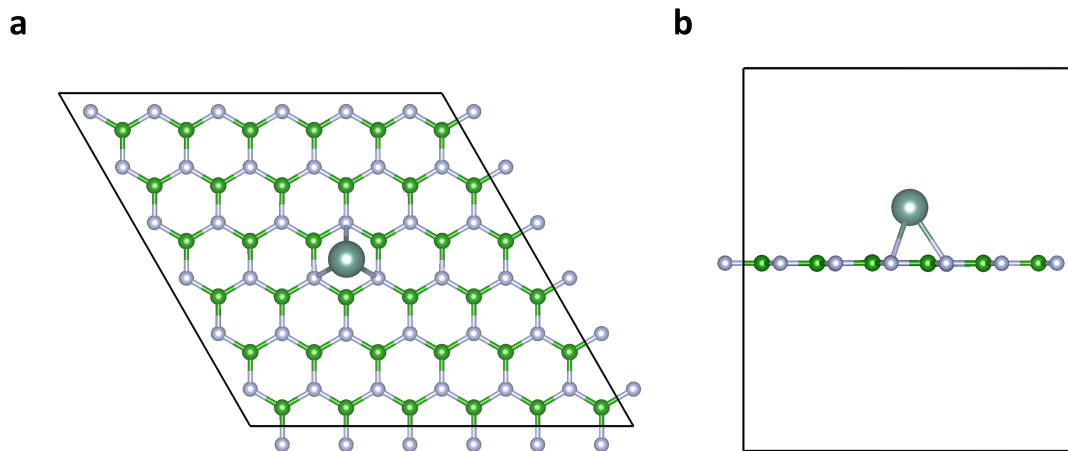
**Figure 7: Stability of single metal atoms supported on graphitic nitride.**



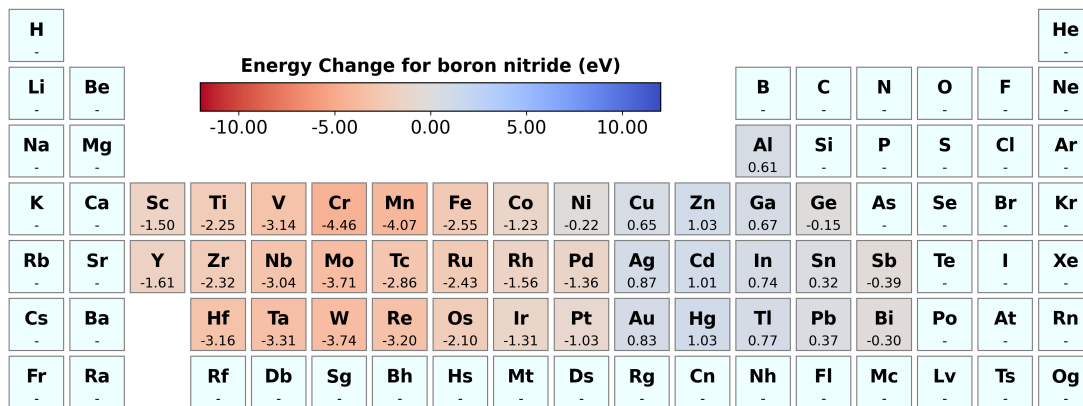
**Figure 8: Structure of single chromium atom supported on black phosphorous.** (a) View along the z-axis and (b) view along the x-axis of the Cr@black phosphorous structure. In the illustrations, light purple spheres represent P atoms, and indigo sphere indicates Cr atom.



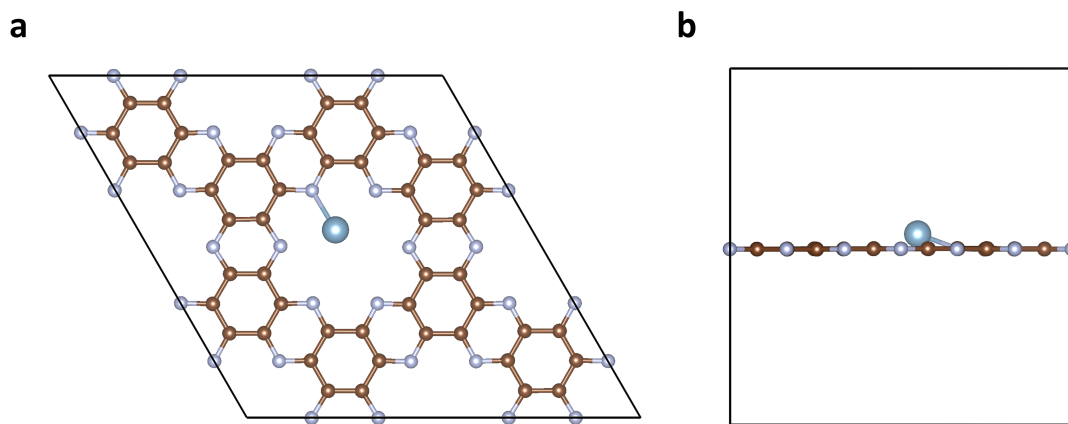
**Figure 9: Stability of single metal atoms supported on graphitic nitride.**



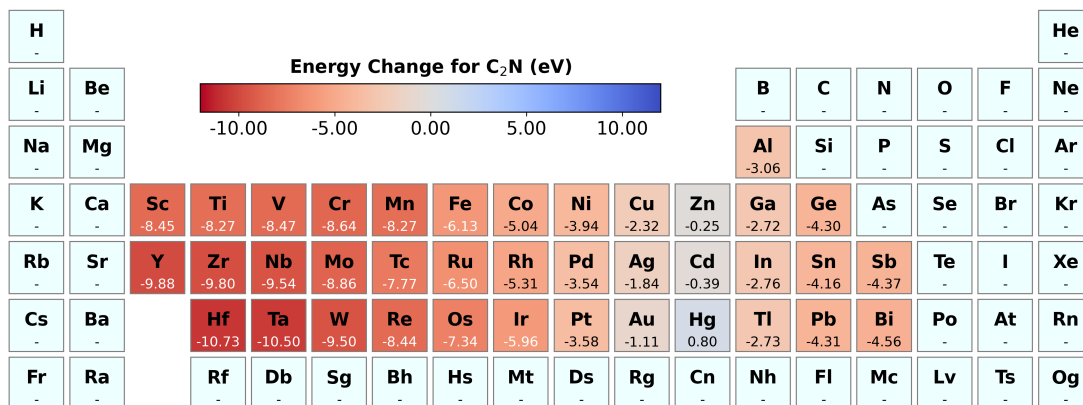
**Figure 10: Structure of single yttrium atom supported on boron nitride.** (a) View along the z-axis and (b) view along the x-axis of the Y@boron-nitride structure. In the illustrations, silver spheres represent N atoms, green spheres signify B atoms, and light green sphere indicates Y atom.



**Figure 11: Stability of single metal atoms supported on graphitic nitride.**



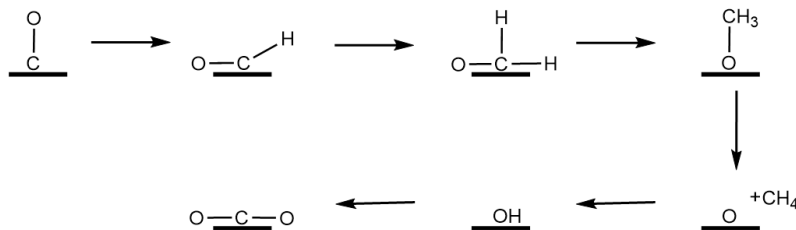
**Figure 12: Structure of single aluminum atom supported on  $C_2N$ .** (a) View along the z-axis and (b) view along the x-axis of the  $Al@C_2N$  structure. In the illustrations, silver spheres represent N atoms, brown spheres signify C atoms, and blue sphere indicates Al atom.



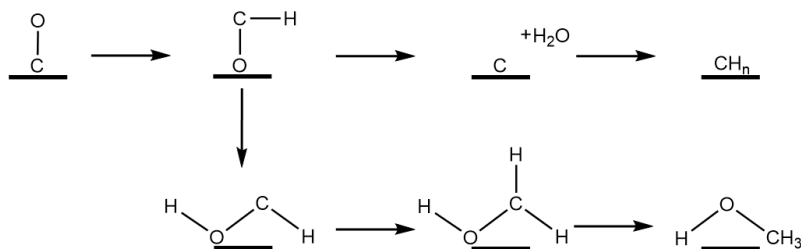
**Figure 13: Stability of single metal atoms supported on graphitic nitride.**

gated. Therefore, to streamline our analysis, we focused exclusively on Mechanism 1 across all catalysts examined.

①



②



③

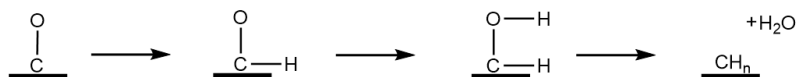


Figure 14: Three investigated CO<sub>2</sub>RR pathways [1, 2, 3] investigated in this work.

### 1.3 DFT energies and energy corrections

### 1.4 Correlation analysis of adsorption energies

Figure 15 presents a Pearson correlation map that illustrates the relationships between the adsorption energies of various intermediates. This analysis encompasses all six substrates under consideration. The mean Pearson correlation coefficient is 0.6023 with a variance of 0.0109, signifying a robust correlation between the adsorption energies of different intermediates, accompanied by low variability.

**Table 1:** DFT-calculated energies in eV for protonated \*CO intermediate for three investigated CO<sub>2</sub>RR pathways.

| Metal | g-C <sub>3</sub> N <sub>4</sub> |           |                  | nitrogen-doped graphene |           |                  | graphene with dual-vacancy |           |                  |
|-------|---------------------------------|-----------|------------------|-------------------------|-----------|------------------|----------------------------|-----------|------------------|
|       | Pathway-1                       | Pathway-2 | Pathway-3        | Pathway-1               | Pathway-2 | Pathway-3        | Pathway-1                  | Pathway-2 | Pathway-3        |
| Al    | <b>-499.3494</b>                | -497.8343 | -499.3005        | <b>-670.3489</b>        | -668.5142 | -670.3093        | -667.2538                  | -666.6725 | <b>-667.2665</b> |
| Co    | -502.5946                       | -501.1257 | <b>-502.6139</b> | -672.8847               | -670.8013 | <b>-672.9020</b> | <b>-671.6284</b>           | -670.4388 | -671.3827        |
| Cr    | -505.6012                       | -504.2022 | <b>-505.6029</b> | -675.3126               | -674.1640 | <b>-675.3480</b> | <b>-673.4283</b>           | -672.1919 | -673.3818        |
| Cu    | <b>-495.8913</b>                | -493.6853 | -495.8820        | <b>-666.9894</b>        | -664.9810 | -666.9705        | -666.9876                  | -664.7684 | <b>-667.1395</b> |
| Fe    | <b>-501.3325</b>                | -499.3969 | -501.3221        | -673.8242               | -672.0887 | <b>-673.8478</b> | <b>-672.7810</b>           | -671.5694 | -672.4960        |
| Ga    | -498.2365                       | -496.3967 | <b>-498.2912</b> | <b>-667.3963</b>        | -665.4677 | -667.3482        | -665.7780                  | -664.8573 | <b>-665.8030</b> |
| Ge    | -500.2052                       | -497.9780 | <b>-500.2156</b> | <b>-668.1785</b>        | -666.3484 | -668.1463        | <b>-667.7387</b>           | -665.7394 | -667.7324        |
| Mn    | <b>-502.6355</b>                | -500.8006 | -502.6294        | -674.6848               | -672.9574 | <b>-674.7030</b> | -673.2243                  | -672.6495 | <b>-673.2361</b> |
| Ni    | <b>-500.4085</b>                | -498.5165 | -500.3985        | <b>-670.0979</b>        | -667.9544 | -670.0832        | -670.1619                  | -668.3684 | <b>-670.1756</b> |
| Sc    | -503.4428                       | -501.6129 | <b>-503.4437</b> | <b>-673.9590</b>        | -671.7624 | -673.3818        | <b>-670.0841</b>           | -668.1044 | -669.6050        |
| Ti    | -503.7848                       | -502.6148 | <b>-503.8033</b> | <b>-675.1006</b>        | -673.2417 | -674.6257        | <b>-672.7127</b>           | -670.6787 | -672.2491        |
| V     | <b>-503.6115</b>                | -502.6188 | -503.5956        | <b>-675.0533</b>        | -674.0394 | -675.0136        | <b>-673.1855</b>           | -671.5382 | -673.1583        |
| Zn    | <b>-493.9854</b>                | -491.5010 | -493.9743        | <b>-664.9935</b>        | -663.0932 | -664.9758        | -663.7735                  | -661.8097 | <b>-663.7959</b> |
| Ag    | <b>-494.4055</b>                | -492.4511 | -494.3905        | -664.6322               | -662.9060 | <b>-664.6353</b> | -664.3972                  | -662.3588 | <b>-665.0570</b> |
| Cd    | <b>-493.3084</b>                | -491.1781 | -493.2951        | <b>-663.8832</b>        | -661.6272 | -663.8653        | -661.8518                  | -659.9601 | <b>-661.8631</b> |
| In    | -498.5272                       | -496.6593 | <b>-498.5908</b> | <b>-666.3968</b>        | -663.7707 | -666.3845        | -663.6532                  | -661.5581 | <b>-663.6913</b> |
| Mo    | <b>-504.5248</b>                | -503.3763 | -504.5148        | -675.3472               | -674.7118 | <b>-675.4389</b> | -674.7186                  | -672.9335 | <b>-674.7397</b> |
| Nb    | -505.3967                       | -504.7113 | <b>-505.3982</b> | <b>-676.0492</b>        | -674.1997 | -675.4107        | <b>-674.6606</b>           | -672.8268 | -674.5488        |
| Pd    | -500.0370                       | -498.3310 | <b>-500.0455</b> | <b>-669.0243</b>        | -667.0759 | -669.0205        | <b>-669.1169</b>           | -667.3223 | -669.1119        |
| Rh    | <b>-499.2225</b>                | -497.3162 | -499.2170        | -672.7920               | -670.7853 | <b>-672.8099</b> | <b>-671.4286</b>           | -670.3036 | -671.4181        |
| Ru    | <b>-503.9986</b>                | -502.1908 | -503.9640        | -674.0273               | -672.4060 | <b>-674.0339</b> | <b>-672.9956</b>           | -672.1332 | -672.9909        |
| Sb    | <b>-497.3836</b>                | -495.1348 | -497.3442        | <b>-666.7198</b>        | -664.9611 | -666.7190        | -666.8632                  | -664.8131 | <b>-666.8670</b> |
| Sn    | <b>-499.4952</b>                | -497.2007 | -499.1116        | <b>-667.1459</b>        | -665.3835 | -667.0999        | <b>-665.9821</b>           | -663.8277 | -665.9755        |
| Tc    | <b>-503.0602</b>                | -501.1754 | -503.0532        | -674.4342               | -673.8164 | <b>-674.4559</b> | -674.0615                  | -672.9747 | <b>-674.0738</b> |
| Y     | <b>-503.9232</b>                | -502.0670 | -503.8988        | <b>-673.7620</b>        | -671.6132 | -673.2241        | -669.4633                  | -668.0403 | <b>-669.5030</b> |
| Zr    | <b>-505.2907</b>                | -503.9148 | -505.1817        | <b>-675.6874</b>        | -673.5456 | -675.0519        | <b>-673.1694</b>           | -671.1993 | -672.8072        |
| Au    | <b>-497.0784</b>                | -495.1720 | -497.0175        | <b>-665.1479</b>        | -663.3239 | -665.1370        | -666.4185                  | -664.2892 | <b>-666.5182</b> |
| Bi    | <b>-496.9946</b>                | -494.7679 | -496.9934        | -665.9563               | -664.2409 | <b>-666.0512</b> | <b>-665.0896</b>           | -663.2974 | -665.0887        |
| Hf    | <b>-506.4837</b>                | -504.9734 | -506.4777        | <b>-677.3077</b>        | -675.1797 | -676.6502        | <b>-674.7209</b>           | -672.6650 | -674.2149        |
| Hg    | <b>-491.0035</b>                | -488.7951 | -490.9728        | <b>-661.2458</b>        | -659.2406 | -661.2398        | -661.7192                  | -659.8398 | <b>-661.7300</b> |
| Ir    | <b>-503.0227</b>                | -501.1368 | -503.0217        | -673.9926               | -672.0068 | <b>-674.0087</b> | <b>-673.5129</b>           | -672.6725 | -673.4975        |
| Os    | -504.7648                       | -502.9878 | <b>-504.8105</b> | -675.2098               | -673.6653 | <b>-675.2235</b> | <b>-674.9483</b>           | -674.1970 | -674.9223        |
| Pb    | <b>-499.1183</b>                | -497.1173 | -499.0544        | -666.4446               | -664.6897 | <b>-666.4625</b> | <b>-664.0200</b>           | -662.2222 | -664.0123        |
| Pt    | <b>-499.3356</b>                | -497.5110 | -499.3215        | -670.3996               | -668.2412 | <b>-670.4147</b> | <b>-671.1408</b>           | -669.2725 | -671.1370        |
| Re    | <b>-504.1830</b>                | -502.4735 | -504.1746        | <b>-675.7670</b>        | -675.0715 | -675.7486        | <b>-675.8701</b>           | -674.9245 | -675.8233        |
| Ta    | -506.6180                       | -505.7793 | <b>-506.6241</b> | <b>-677.7222</b>        | -675.9907 | -677.0911        | <b>-676.4371</b>           | -674.5434 | -676.4262        |
| Tl    | <b>-497.9559</b>                | -496.1578 | -497.9382        | -664.5047               | -662.7051 | <b>-664.5051</b> | -662.6279                  | -660.6086 | <b>-662.6317</b> |
| W     | <b>-505.8134</b>                | -504.9085 | -505.8030        | -676.5304               | -676.1384 | <b>-676.9965</b> | <b>-676.7669</b>           | -675.4895 | -676.7356        |

*Note:* Energies of the most stable configurations are highlighted in bold.

**Table 2:** Free energies at 298.15 K for isolated species.

| Species           | Free Energy (eV) |
|-------------------|------------------|
| CO <sub>2</sub>   | -23.3140         |
| CO                | -15.3336         |
| CH <sub>4</sub>   | -23.4639         |
| H <sub>2</sub>    | -6.9315          |
| H <sub>2</sub> O  | -14.3239         |
| COOH              | -24.3963         |
| CHO               | -17.0983         |
| CH <sub>2</sub> O | -22.1607         |
| OCH <sub>3</sub>  | -24.5075         |
| O                 | -1.9007          |
| OH                | -7.7303          |

*Note:* VASPKIT [4] was employed to calculate free energies.

**Table 3:** Free energy corrections at 298.15 K for adsorbed intermediates.

| Species            | Free energy correction (eV) |
|--------------------|-----------------------------|
| *CO <sub>2</sub>   | 0.0864                      |
| *COOH              | 0.3693                      |
| *CO                | 0.0111                      |
| *CHO               | 0.2716                      |
| *CH <sub>2</sub> O | 0.5595                      |
| *OCH <sub>3</sub>  | 0.8738                      |
| *O                 | -0.0193                     |
| *OH                | 0.1993                      |
| *H                 | 0.1533                      |

*Note:* Free energy corrections were obtained by averaging the DFT-computed corrections, which include zero-point energy and entropy, across three substrates: g-C<sub>3</sub>N<sub>4</sub>, nitrogen-doped graphene, and dual-vacancy graphene.

**Table 4:** DFT calculated final state energies in eV for adsorbates supported on g-C<sub>3</sub>N<sub>4</sub>.

| Metal | *CO <sub>2</sub> | *COOH     | *CO       | *CHO      | *CH <sub>2</sub> O | *OCH <sub>3</sub> | *O        | *OH       | *H        |
|-------|------------------|-----------|-----------|-----------|--------------------|-------------------|-----------|-----------|-----------|
| Al    | -481.2292        | -504.7972 | -508.6897 | -496.4490 | -500.7640          | -503.7935         | -509.9998 | -488.6623 | -485.4855 |
| Co    | -483.2039        | -507.0765 | -510.0453 | -500.1329 | -503.9316          | -507.0552         | -510.9052 | -489.9487 | -486.6194 |
| Cr    | -486.7813        | -509.9599 | -513.3962 | -502.1493 | -505.7125          | -509.8851         | -514.1702 | -493.7457 | -489.9202 |
| Cu    | -477.5955        | -502.5914 | -505.9613 | -495.6511 | -498.0502          | -502.4590         | -506.2829 | -484.3166 | -481.9683 |
| Fe    | -482.0576        | -507.1508 | -510.2918 | -500.1101 | -502.1626          | -506.0115         | -511.2543 | -490.0116 | -486.4067 |
| Ga    | -480.8854        | -503.9317 | -506.4938 | -495.7379 | -498.7813          | -503.1345         | -507.5029 | -486.1702 | -483.4575 |
| Ge    | -481.5807        | -504.6467 | -508.4204 | -496.4421 | -500.5712          | -504.0218         | -509.4116 | -488.4168 | -485.2680 |
| Mn    | -483.7690        | -508.5410 | -511.6514 | -500.6279 | -502.6739          | -508.6433         | -512.8569 | -491.2239 | -488.1082 |
| Ni    | -482.0061        | -505.1988 | -508.4173 | -498.5054 | -501.0040          | -504.8945         | -508.6266 | -487.3684 | -484.8272 |
| Sc    | -483.9963        | -509.3147 | -512.9882 | -501.4505 | -505.2024          | -509.4054         | -514.5969 | -493.4105 | -489.2632 |
| Ti    | -484.2282        | -509.6510 | -512.8528 | -502.0410 | -505.3761          | -509.9729         | -514.0700 | -493.6478 | -489.2234 |
| V     | -484.1990        | -508.6926 | -512.3009 | -501.6777 | -504.4297          | -509.3618         | -513.4561 | -493.8569 | -488.9474 |
| Zn    | -475.2375        | -500.3556 | -504.4103 | -492.4144 | -496.5426          | -499.4232         | -504.6297 | -482.6782 | -481.0438 |
| Ag    | -476.7049        | -501.6430 | -504.5861 | -493.9138 | -496.7220          | -500.9477         | -504.5408 | -482.6604 | -481.2659 |
| Cd    | -475.1288        | -500.1854 | -503.4108 | -491.8954 | -495.6486          | -499.4959         | -503.6209 | -483.3427 | -479.8278 |
| In    | -481.1330        | -504.1653 | -506.4422 | -496.0016 | -498.6643          | -503.3286         | -507.3780 | -485.6281 | -483.1445 |
| Mo    | -484.2892        | -509.4263 | -512.9103 | -503.0025 | -505.5534          | -509.6730         | -513.0136 | -493.5700 | -489.4994 |
| Nb    | -485.4055        | -510.7952 | -514.0380 | -503.5663 | -506.3671          | -511.1572         | -514.7845 | -495.5118 | -490.2111 |
| Pd    | -481.5399        | -504.7593 | -508.5173 | -498.0817 | -501.1620          | -504.0892         | -508.1976 | -486.1551 | -483.9553 |
| Rh    | -479.8884        | -505.2422 | -508.6754 | -498.7129 | -501.4594          | -504.5254         | -508.9536 | -487.8771 | -485.0738 |
| Ru    | -483.8526        | -507.5120 | -511.5881 | -500.6857 | -504.1497          | -506.8934         | -510.9524 | -490.6690 | -488.0169 |
| Sb    | -478.5169        | -502.9708 | -506.5211 | -494.8740 | -498.9453          | -502.5347         | -507.2111 | -486.4448 | -483.4418 |
| Sn    | -481.4238        | -504.4677 | -507.8470 | -496.2883 | -499.9258          | -503.6685         | -508.8152 | -487.6963 | -484.6351 |
| Tc    | -482.7091        | -507.9652 | -512.1064 | -501.9896 | -504.6034          | -508.3486         | -511.4777 | -492.0381 | -488.0897 |
| Y     | -484.6612        | -509.8564 | -513.6851 | -501.9522 | -505.8238          | -509.8344         | -515.2400 | -493.4038 | -489.6506 |
| Zr    | -485.4694        | -511.1852 | -514.3901 | -503.1958 | -506.9346          | -511.5864         | -515.6720 | -496.3037 | -490.4785 |
| Au    | -479.1941        | -502.3245 | -506.9013 | -495.6782 | -499.0711          | -501.6877         | -506.5462 | -484.6198 | -483.6909 |
| Bi    | -478.2695        | -502.8490 | -506.2055 | -494.6965 | -498.5557          | -502.3398         | -506.7999 | -485.7536 | -483.1348 |
| Hf    | -486.5155        | -512.4535 | -515.5965 | -504.3393 | -508.1599          | -512.8332         | -517.2187 | -496.4195 | -491.7486 |
| Hg    | -472.8730        | -500.0491 | -502.1889 | -491.7544 | -494.1224          | -499.2181         | -501.8365 | -480.3761 | -479.2046 |
| Ir    | -483.7878        | -507.2024 | -510.6763 | -500.2051 | -503.2948          | -506.5209         | -509.8509 | -490.9790 | -487.3503 |
| Os    | -484.9279        | -508.6427 | -512.5954 | -501.8541 | -505.2080          | -507.8334         | -511.8734 | -491.0155 | -489.0898 |
| Pb    | -481.3587        | -504.4450 | -507.5634 | -496.2379 | -499.6114          | -503.5822         | -508.1698 | -486.8476 | -484.2646 |
| Pt    | -481.1676        | -504.3670 | -508.2170 | -499.0073 | -500.9446          | -504.2695         | -508.1465 | -487.3508 | -484.6644 |
| Re    | -483.6421        | -508.9474 | -512.8388 | -502.8952 | -505.3651          | -509.1168         | -512.6374 | -493.0621 | -489.2811 |
| Ta    | -486.3299        | -512.1371 | -515.1895 | -504.6585 | -507.5888          | -512.5295         | -516.0165 | -497.0134 | -490.5695 |
| Tl    | -480.7357        | -503.7632 | -505.9937 | -495.5894 | -497.7724          | -503.0564         | -506.7919 | -484.0732 | -482.3846 |
| W     | -485.2596        | -510.8145 | -514.28   | -504.2714 | -506.7988          | -511.0643         | -514.2358 | -494.4330 | -490.6506 |



**Table 5:** DFT calculated final state energies in eV for adsorbates supported on nitrogen-doped graphene.

| Metal | *CO <sub>2</sub> | *COOH     | *CO       | *CHO      | *CH <sub>2</sub> O | *OCH <sub>3</sub> | *O        | *OH       | *H        |
|-------|------------------|-----------|-----------|-----------|--------------------|-------------------|-----------|-----------|-----------|
| Al    | -652.7589        | -675.7997 | -679.7595 | -668.0296 | -671.7596          | -676.2161         | -681.1311 | -658.7849 | -656.4702 |
| Co    | -654.3280        | -677.5394 | -681.2518 | -670.0860 | -673.9208          | -676.7761         | -680.7219 | -659.1714 | -657.8777 |
| Cr    | -657.0479        | -680.2074 | -683.7982 | -673.0760 | -676.1859          | -679.4701         | -684.5295 | -664.7306 | -660.4141 |
| Cu    | -650.2181        | -673.3928 | -675.5873 | -665.2174 | -668.0662          | -672.5797         | -675.7047 | -653.5770 | -652.1010 |
| Fe    | -655.4034        | -678.5897 | -682.1186 | -671.9101 | -674.7438          | -677.8515         | -682.1216 | -661.5422 | -658.7383 |
| Ga    | -649.9563        | -672.9807 | -676.9763 | -664.9727 | -669.0739          | -672.8536         | -677.7169 | -655.5345 | -653.9243 |
| Ge    | -651.0391        | -674.1655 | -677.2582 | -665.9255 | -669.5282          | -673.3438         | -678.1585 | -657.4964 | -654.3444 |
| Mn    | -656.5635        | -679.7128 | -683.1092 | -672.6753 | -675.6877          | -679.0710         | -683.3898 | -663.1584 | -659.7760 |
| Ni    | -653.1461        | -676.3270 | -678.5166 | -668.1448 | -671.1502          | -675.5454         | -678.5411 | -656.3929 | -655.1238 |
| Sc    | -655.2274        | -678.9985 | -682.8810 | -670.9489 | -674.9850          | -679.1986         | -684.3192 | -662.2442 | -659.0243 |
| Ti    | -655.5937        | -680.2412 | -683.5514 | -671.9689 | -675.9724          | -680.4589         | -685.1366 | -665.2871 | -659.7132 |
| V     | -656.0751        | -680.2584 | -683.5739 | -672.6281 | -676.0229          | -680.5926         | -684.9400 | -665.6038 | -659.9017 |
| Zn    | -648.2316        | -671.3824 | -674.1973 | -663.2623 | -666.3990          | -670.5613         | -674.5436 | -652.2900 | -650.9227 |
| Ag    | -646.9759        | -670.2666 | -673.2766 | -663.1058 | -665.7601          | -669.7801         | -673.0113 | -651.0760 | -649.9287 |
| Cd    | -646.1687        | -669.2620 | -672.8043 | -661.4439 | -664.9473          | -668.8180         | -672.9588 | -650.7684 | -649.7670 |
| In    | -648.6726        | -671.7218 | -675.3788 | -663.4885 | -667.5183          | -670.7536         | -675.8630 | -653.8535 | -652.4757 |
| Mo    | -655.7944        | -680.2966 | -683.6262 | -672.7667 | -676.3955          | -680.6291         | -684.8234 | -665.8138 | -659.8169 |
| Nb    | -656.0277        | -680.8814 | -684.2662 | -672.5576 | -676.9139          | -681.4203         | -685.7720 | -666.2363 | -660.3734 |
| Pd    | -652.2091        | -675.3967 | -677.2760 | -667.1855 | -669.8609          | -674.6033         | -677.3498 | -655.1080 | -654.0027 |
| Rh    | -653.6928        | -676.9529 | -680.9755 | -669.5137 | -673.6394          | -676.2879         | -680.2104 | -658.6953 | -657.6908 |
| Ru    | -654.5425        | -678.2124 | -681.9561 | -672.0589 | -674.6177          | -678.0945         | -681.9008 | -661.3838 | -658.6572 |
| Sb    | -649.9262        | -672.9182 | -675.6457 | -664.7128 | -667.9159          | -672.0849         | -675.8420 | -655.1699 | -652.4240 |
| Sn    | -650.5072        | -673.6118 | -675.7767 | -665.3842 | -668.0544          | -672.7597         | -676.5046 | -655.8947 | -652.9261 |
| Tc    | -655.4025        | -679.3044 | -682.7454 | -672.5785 | -676.3294          | -679.4840         | -683.3209 | -664.2323 | -659.2703 |
| Y     | -655.2660        | -679.0430 | -682.7571 | -670.8208 | -674.7780          | -679.0775         | -684.1210 | -661.8034 | -658.8885 |
| Zr    | -655.9118        | -680.6859 | -684.1920 | -672.1497 | -676.4014          | -680.7930         | -685.7940 | -665.4878 | -660.4163 |
| Au    | -648.3344        | -671.3969 | -673.1469 | -663.1955 | -665.5964          | -670.5800         | -673.6092 | -651.2436 | -649.7275 |
| Bi    | -649.1484        | -672.1225 | -675.1630 | -663.9594 | -667.2981          | -671.4617         | -675.6926 | -654.7684 | -651.8907 |
| Hf    | -657.3364        | -682.3019 | -685.8119 | -673.6603 | -678.0090          | -682.4001         | -687.4607 | -666.9550 | -662.0651 |
| Hg    | -644.7368        | -667.9224 | -671.2200 | -659.6906 | -663.5208          | -667.0466         | -670.9016 | -648.8944 | -648.3755 |
| Ir    | -654.8564        | -678.1079 | -682.2865 | -670.9826 | -674.9484          | -677.4097         | -681.3088 | -660.1543 | -658.9895 |
| Os    | -655.6722        | -679.5364 | -683.3240 | -673.4289 | -675.9254          | -679.4423         | -683.1300 | -663.0791 | -659.9901 |
| Pb    | -649.8843        | -672.9588 | -674.5856 | -664.7553 | -667.1174          | -672.0886         | -675.8823 | -653.6827 | -651.5940 |
| Pt    | -653.3281        | -676.5221 | -678.6094 | -668.2951 | -671.2463          | -675.7317         | -678.4334 | -656.4419 | -655.4733 |
| Re    | -656.3492        | -680.7664 | -683.9596 | -673.9732 | -676.5164          | -681.0627         | -684.8096 | -665.9526 | -660.5620 |
| Ta    | -657.1988        | -682.6048 | -685.8231 | -674.1666 | -678.5029          | -682.9709         | -687.2294 | -667.7321 | -661.8794 |
| Tl    | -648.1346        | -671.1831 | -673.2953 | -662.9610 | -665.6240          | -670.3512         | -673.4567 | -651.7909 | -650.4750 |
| W     | -657.0204        | -682.0186 | -685.3735 | -674.3843 | -678.2483          | -682.4377         | -686.3613 | -667.4547 | -661.5686 |

**Table 6:** DFT calculated final state energies in eV for adsorbates supported on graphene with dual-vacancy.

| Metal | *CO <sub>2</sub> | *COOH     | *CO       | *CHO      | *CH <sub>2</sub> O | *OCH <sub>3</sub> | *O        | *OH       | *H        |
|-------|------------------|-----------|-----------|-----------|--------------------|-------------------|-----------|-----------|-----------|
| Al    | -650.3496        | -673.3468 | -676.2273 | -665.6303 | -668.4112          | -672.5763         | -677.4453 | -655.0864 | -653.0166 |
| Co    | -652.5157        | -675.5864 | -679.3861 | -669.0614 | -672.8955          | -675.8508         | -679.5268 | -658.7929 | -655.9377 |
| Cr    | -655.0421        | -678.1256 | -681.8470 | -671.1282 | -674.2868          | -679.1554         | -683.4111 | -663.3554 | -658.4913 |
| Cu    | -649.8187        | -672.9161 | -674.9327 | -664.8190 | -669.0819          | -672.1515         | -674.7438 | -652.8789 | -653.6953 |
| Fe    | -653.7964        | -677.0127 | -680.3412 | -670.0219 | -673.9525          | -676.7631         | -681.1346 | -660.9239 | -656.9759 |
| Ga    | -649.0340        | -672.0433 | -674.9630 | -663.8505 | -667.2101          | -671.2541         | -675.5229 | -653.2086 | -651.8693 |
| Ge    | -651.0133        | -674.2532 | -677.2272 | -666.0554 | -669.5660          | -673.4485         | -677.8152 | -655.7843 | -654.2731 |
| Mn    | -654.8183        | -678.0584 | -681.5834 | -670.9593 | -674.0994          | -677.9260         | -682.6048 | -662.5702 | -658.1064 |
| Ni    | -651.5295        | -674.6692 | -677.8941 | -667.2937 | -670.7325          | -673.9392         | -677.5204 | -656.1026 | -655.6559 |
| Sc    | -652.1630        | -675.5777 | -678.6101 | -667.5024 | -670.7255          | -675.4361         | -680.0287 | -658.0834 | -654.6913 |
| Ti    | -654.2548        | -677.9326 | -681.1516 | -670.0780 | -673.2830          | -677.9113         | -682.6933 | -661.1957 | -657.4040 |
| V     | -654.6939        | -678.0200 | -681.8606 | -670.8517 | -674.2298          | -679.2097         | -683.4928 | -663.1639 | -658.4619 |
| Zn    | -646.9805        | -670.1676 | -672.2508 | -662.0849 | -664.7199          | -669.3791         | -672.6028 | -650.3134 | -648.8961 |
| Ag    | -647.4840        | -670.5954 | -672.4765 | -662.3757 | -665.1014          | -669.8199         | -672.2893 | -650.2207 | -651.5223 |
| Cd    | -645.2038        | -668.3897 | -670.2685 | -660.2502 | -662.7479          | -667.5994         | -670.6023 | -648.2958 | -646.9658 |
| In    | -646.8189        | -669.9379 | -673.1820 | -661.7351 | -665.3833          | -669.1524         | -673.5557 | -651.3730 | -650.2004 |
| Mo    | -656.2058        | -679.7202 | -683.2424 | -672.3832 | -675.6459          | -679.9903         | -684.8002 | -664.6331 | -659.9673 |
| Nb    | -656.1010        | -679.4842 | -683.3623 | -672.0587 | -675.6683          | -679.5996         | -684.8935 | -664.2634 | -659.9617 |
| Pd    | -650.5405        | -673.6524 | -677.0119 | -666.0816 | -669.7529          | -672.9394         | -676.5341 | -655.1586 | -654.8764 |
| Rh    | -652.7998        | -675.8992 | -679.6974 | -669.1468 | -672.2062          | -676.2142         | -679.9592 | -659.4214 | -656.7388 |
| Ru    | -654.8166        | -678.0140 | -681.6090 | -671.1415 | -674.0502          | -677.9828         | -682.4415 | -662.4653 | -658.2092 |
| Sb    | -649.9882        | -673.0639 | -675.9906 | -664.8414 | -668.3621          | -672.2685         | -676.6274 | -655.1055 | -653.1345 |
| Sn    | -648.5997        | -671.7246 | -675.2765 | -663.7120 | -667.5399          | -670.8850         | -675.8484 | -653.8138 | -652.4041 |
| Tc    | -655.8173        | -679.2476 | -682.6345 | -672.1709 | -675.0076          | -679.2729         | -683.7569 | -664.1434 | -659.2908 |
| Y     | -652.1517        | -675.4891 | -678.6682 | -667.4644 | -670.6934          | -675.1478         | -680.0013 | -658.3045 | -654.7163 |
| Zr    | -654.8760        | -678.4572 | -681.6839 | -670.6090 | -673.8291          | -678.4537         | -683.1390 | -661.4741 | -657.9892 |
| Au    | -649.3963        | -672.5116 | -674.3660 | -664.2836 | -667.3357          | -671.7273         | -674.1359 | -652.0358 | -653.1138 |
| Bi    | -648.8651        | -672.0004 | -673.7920 | -663.7934 | -666.2170          | -671.1801         | -674.0672 | -652.8628 | -650.9149 |
| Hf    | -656.1155        | -679.7805 | -683.2130 | -671.9925 | -675.3123          | -679.9256         | -684.7502 | -662.8630 | -659.4860 |
| Hg    | -645.0948        | -668.2801 | -669.9749 | -660.1062 | -662.5696          | -667.4822         | -670.2672 | -647.8582 | -646.7652 |
| Ir    | -654.4999        | -677.8310 | -681.7312 | -671.1936 | -674.7821          | -678.1749         | -681.9532 | -661.9662 | -658.5262 |
| Os    | -656.3635        | -679.8433 | -683.4557 | -673.1036 | -675.9479          | -679.6710         | -684.4704 | -664.7281 | -660.0873 |
| Pb    | -647.8007        | -670.9491 | -673.2996 | -662.7392 | -665.6502          | -670.1373         | -673.5969 | -651.6692 | -650.4432 |
| Pt    | -652.4961        | -675.6092 | -678.9183 | -668.2053 | -671.6867          | -674.8876         | -678.4355 | -657.4936 | -656.2759 |
| Re    | -657.2440        | -681.0777 | -684.5183 | -673.9603 | -676.8950          | -681.2383         | -685.8328 | -666.1615 | -661.2885 |
| Ta    | -657.4068        | -681.0711 | -685.1062 | -673.5947 | -677.3738          | -681.6985         | -686.7561 | -666.0327 | -661.7361 |
| Tl    | -645.7286        | -668.8734 | -671.7205 | -660.7259 | -664.0001          | -668.1110         | -672.0305 | -649.7198 | -648.7563 |
| W     | -657.5901        | -681.6722 | -685.1010 | -674.1028 | -677.7828          | -682.4493         | -686.6385 | -666.9364 | -661.8641 |

**Table 7:** Zero-point energies in eV for relaxed adsorbates supported on g-C<sub>3</sub>N<sub>4</sub> at 298.15 K.

| Metal | *CO <sub>2</sub> | *COOH  | *CO    | *CHO   | *CH <sub>2</sub> O | *OCH <sub>3</sub> | *O     | *OH    | *H     |
|-------|------------------|--------|--------|--------|--------------------|-------------------|--------|--------|--------|
| Al    | 0.2905           | 0.6130 | 0.1743 | 0.4606 | 0.7456             | 1.0974            | 0.0629 | 0.3521 | 0.1925 |
| Co    | 0.3038           | 0.5895 | 0.2051 | 0.4186 | 0.6269             | 1.0663            | 0.0571 | 0.3248 | 0.1507 |
| Cr    | 0.3140           | 0.5964 | 0.1640 | 0.4374 | 0.7725             | 1.0779            | 0.0670 | 0.3286 | 0.1626 |
| Cu    | 0.3157           | 0.6016 | 0.2045 | 0.4498 | 0.7797             | 1.0751            | 0.0494 | 0.3332 | 0.1577 |
| Fe    | 0.3088           | 0.5859 | 0.1931 | 0.4437 | 0.7720             | 1.0726            | 0.0599 | 0.3326 | 0.1550 |
| Ga    | 0.3102           | 0.6131 | 0.1448 | 0.4665 | 0.7186             | 1.0680            | 0.0608 | 0.3202 | 0.1813 |
| Ge    | 0.3091           | 0.6069 | 0.1396 | 0.4540 | 0.7269             | 1.0881            | 0.0581 | 0.3506 | 0.1904 |
| Mn    | 0.3022           | 0.5876 | 0.1733 | 0.4333 | 0.7637             | 1.0622            | 0.0687 | 0.3219 | 0.1432 |
| Ni    | 0.3023           | 0.5934 | 0.2074 | 0.4546 | 0.7662             | 1.0623            | 0.0542 | 0.3313 | 0.1611 |
| Sc    | 0.2880           | 0.6039 | 0.1771 | 0.4544 | 0.7473             | 1.0806            | 0.0657 | 0.3156 | 0.1480 |
| Ti    | 0.3044           | 0.6088 | 0.1889 | 0.4664 | 0.7863             | 1.0862            | 0.0796 | 0.3417 | 0.1590 |
| V     | 0.3081           | 0.6005 | 0.2031 | 0.4408 | 0.7868             | 1.0882            | 0.0792 | 0.3388 | 0.1660 |
| Zn    | 0.3243           | 0.6080 | 0.1729 | 0.4601 | 0.7277             | 1.0832            | 0.0514 | 0.3468 | 0.1726 |
| Ag    | 0.3167           | 0.5993 | 0.1815 | 0.4457 | 0.7264             | 1.0590            | 0.0390 | 0.3304 | 0.1576 |
| Cd    | 0.3242           | 0.6007 | 0.1533 | 0.4502 | 0.7292             | 1.0604            | 0.0983 | 0.3322 | 0.1439 |
| In    | 0.3116           | 0.5687 | 0.1399 | 0.4025 | 0.7158             | 1.0551            | 0.0523 | 0.3315 | 0.1141 |
| Mo    | 0.3109           | 0.6071 | 0.2163 | 0.4441 | 0.7970             | 1.0825            | 0.0765 | 0.3312 | 0.1851 |
| Nb    | 0.3101           | 0.5971 | 0.2013 | 0.4340 | 0.7960             | 1.0908            | 0.0798 | 0.3329 | 0.1598 |
| Pd    | 0.3148           | 0.6091 | 0.1996 | 0.4778 | 0.7402             | 1.0777            | 0.0525 | 0.3425 | 0.1508 |
| Rh    | 0.3064           | 0.6180 | 0.2215 | 0.4654 | 0.7649             | 1.0844            | 0.0615 | 0.3643 | 0.1946 |
| Ru    | 0.3060           | 0.6201 | 0.2104 | 0.4668 | 0.7761             | 1.0800            | 0.0663 | 0.3421 | 0.2026 |
| Sb    | 0.3050           | 0.6244 | 0.1461 | 0.4696 | 0.7473             | 1.1052            | 0.0689 | 0.3676 | 0.2043 |
| Sn    | 0.3115           | 0.5941 | 0.1441 | 0.4373 | 0.7202             | 1.0761            | 0.0591 | 0.3397 | 0.1601 |
| Tc    | 0.3170           | 0.6102 | 0.2178 | 0.4543 | 0.7890             | 1.0807            | 0.0698 | 0.3442 | 0.2008 |
| Y     | 0.2768           | 0.5949 | 0.1712 | 0.4467 | 0.7278             | 1.0780            | 0.0573 | 0.3206 | 0.1349 |
| Zr    | 0.2939           | 0.6068 | 0.1888 | 0.4631 | 0.7737             | 1.0875            | 0.0744 | 0.3155 | 0.1592 |
| Au    | 0.3123           | 0.6245 | 0.2147 | 0.4761 | 0.7312             | 1.0845            | 0.0527 | 0.3432 | 0.2066 |
| Bi    | 0.3076           | 0.6148 | 0.1485 | 0.4612 | 0.7392             | 1.0935            | 0.0611 | 0.3554 | 0.1893 |
| Hf    | 0.2928           | 0.6104 | 0.1880 | 0.4658 | 0.7769             | 1.0954            | 0.0765 | 0.3218 | 0.1728 |
| Hg    | 0.3220           | 0.6221 | 0.1486 | 0.4707 | 0.7338             | 1.0809            | 0.0541 | 0.3469 | 0.1924 |
| Ir    | 0.3127           | 0.6311 | 0.2271 | 0.4818 | 0.7530             | 1.0740            | 0.1084 | 0.3314 | 0.2142 |
| Os    | 0.3122           | 0.6260 | 0.2258 | 0.4796 | 0.7531             | 1.0911            | 0.0670 | 0.3501 | 0.2154 |
| Pb    | 0.3152           | 0.5834 | 0.1401 | 0.4239 | 0.7113             | 1.0486            | 0.0513 | 0.3043 | 0.1352 |
| Pt    | 0.3144           | 0.6361 | 0.2225 | 0.4908 | 0.7569             | 1.0921            | 0.0603 | 0.3672 | 0.2058 |
| Re    | 0.3171           | 0.6175 | 0.2213 | 0.4665 | 0.8021             | 1.0985            | 0.0780 | 0.3597 | 0.2032 |
| Ta    | 0.3099           | 0.5977 | 0.2000 | 0.4414 | 0.7993             | 1.0880            | 0.0847 | 0.3410 | 0.1780 |
| Tl    | 0.3114           | 0.5614 | 0.1389 | 0.3494 | 0.7105             | 1.0410            | 0.0246 | 0.3210 | 0.0951 |
| W     | 0.3146           | 0.6059 | 0.2160 | 0.4456 | 0.8033             | 1.0886            | 0.0725 | 0.3387 | 0.1834 |

**Table 8:** Entropy corrections ( $-T \cdot S$ ) in eV for relaxed adsorbates supported on g-C<sub>3</sub>N<sub>4</sub> at 298.15 K.

| Metal | *CO <sub>2</sub> | *COOH   | *CO     | *CHO    | *CH <sub>2</sub> O | *OCH <sub>3</sub> | *O      | *OH     | *H      |
|-------|------------------|---------|---------|---------|--------------------|-------------------|---------|---------|---------|
| Al    | -0.2195          | -0.2476 | -0.1874 | -0.1939 | -0.2359            | -0.1759           | -0.0894 | -0.1010 | -0.0124 |
| Co    | -0.2359          | -0.2665 | -0.1544 | -0.1572 | -0.1755            | -0.2095           | -0.1078 | -0.1510 | -0.0341 |
| Cr    | -0.1862          | -0.2605 | -0.2075 | -0.2069 | -0.1478            | -0.1733           | -0.0790 | -0.1437 | -0.0221 |
| Cu    | -0.3098          | -0.2403 | -0.1655 | -0.1803 | -0.1647            | -0.2378           | -0.0977 | -0.1375 | -0.0297 |
| Fe    | -0.2184          | -0.2028 | -0.1676 | -0.1797 | -0.1742            | -0.2640           | -0.0815 | -0.1285 | -0.0215 |
| Ga    | -0.2602          | -0.2558 | -0.2291 | -0.1881 | -0.1252            | -0.2576           | -0.0897 | -0.1441 | -0.0182 |
| Ge    | -0.1978          | -0.2423 | -0.1635 | -0.1802 | -0.2104            | -0.2364           | -0.1178 | -0.1052 | -0.0087 |
| Mn    | -0.2142          | -0.2678 | -0.1863 | -0.2059 | -0.1788            | -0.2752           | -0.0808 | -0.1425 | -0.0261 |
| Ni    | -0.2460          | -0.1948 | -0.1592 | -0.1224 | -0.1275            | -0.2506           | -0.0997 | -0.1384 | -0.0224 |
| Sc    | -0.2227          | -0.2369 | -0.1843 | -0.1667 | -0.1817            | -0.2035           | -0.0728 | -0.1416 | -0.0226 |
| Ti    | -0.1978          | -0.2263 | -0.1685 | -0.1389 | -0.1414            | -0.1795           | -0.0578 | -0.1013 | -0.0195 |
| V     | -0.2603          | -0.2511 | -0.1452 | -0.2028 | -0.1360            | -0.2374           | -0.0612 | -0.1082 | -0.0186 |
| Zn    | -0.2928          | -0.2543 | -0.1909 | -0.1959 | -0.2202            | -0.2331           | -0.0910 | -0.1185 | -0.0228 |
| Ag    | -0.2458          | -0.2428 | -0.1910 | -0.1890 | -0.3097            | -0.2421           | -0.1060 | -0.1392 | -0.0276 |
| Cd    | -0.2961          | -0.2561 | -0.2368 | -0.2023 | -0.2076            | -0.2630           | -0.0270 | -0.1408 | -0.0337 |
| In    | -0.1978          | -0.2895 | -0.1071 | -0.1840 | -0.1338            | -0.2826           | -0.1011 | -0.1386 | -0.0405 |
| Mo    | -0.1912          | -0.2310 | -0.1310 | -0.1680 | -0.1269            | -0.2228           | -0.0641 | -0.1496 | -0.0155 |
| Nb    | -0.1896          | -0.2540 | -0.1460 | -0.1924 | -0.1290            | -0.2147           | -0.0575 | -0.1177 | -0.0261 |
| Pd    | -0.2405          | -0.2385 | -0.1649 | -0.1740 | -0.2538            | -0.2201           | -0.0900 | -0.1143 | -0.0518 |
| Rh    | -0.2408          | -0.2385 | -0.1355 | -0.1734 | -0.1853            | -0.2247           | -0.0863 | -0.0866 | -0.0222 |
| Ru    | -0.2197          | -0.2304 | -0.1450 | -0.1769 | -0.1655            | -0.1574           | -0.0832 | -0.1294 | -0.0138 |
| Sb    | -0.2354          | -0.2277 | -0.1320 | -0.1793 | -0.2211            | -0.1961           | -0.0616 | -0.0808 | -0.0090 |
| Sn    | -0.2606          | -0.2544 | -0.2227 | -0.2021 | -0.3045            | -0.2568           | -0.0787 | -0.1171 | -0.0160 |
| Tc    | -0.1875          | -0.2373 | -0.1347 | -0.1733 | -0.1485            | -0.2301           | -0.0697 | -0.1007 | -0.0121 |
| Y     | -0.2557          | -0.2528 | -0.1929 | -0.1886 | -0.2130            | -0.2081           | -0.0824 | -0.1393 | -0.0285 |
| Zr    | -0.2083          | -0.2325 | -0.1624 | -0.1459 | -0.1574            | -0.2014           | -0.0640 | -0.1556 | -0.0208 |
| Au    | -0.1901          | -0.2267 | -0.1484 | -0.1846 | -0.2939            | -0.2307           | -0.0857 | -0.1271 | -0.0138 |
| Bi    | -0.2366          | -0.2392 | -0.1915 | -0.1899 | -0.2361            | -0.1986           | -0.0683 | -0.0934 | -0.0100 |
| Hf    | -0.2043          | -0.2219 | -0.1618 | -0.1418 | -0.1489            | -0.1981           | -0.0610 | -0.1414 | -0.0163 |
| Hg    | -0.2980          | -0.2371 | -0.2812 | -0.1995 | -0.3039            | -0.2465           | -0.0878 | -0.1245 | -0.0175 |
| Ir    | -0.2097          | -0.2290 | -0.1261 | -0.1803 | -0.1913            | -0.2221           | -0.0377 | -0.1379 | -0.0159 |
| Os    | -0.1795          | -0.2235 | -0.1294 | -0.1519 | -0.1609            | -0.2027           | -0.0680 | -0.1093 | -0.0124 |
| Pb    | -0.2546          | -0.2691 | -0.1666 | -0.2165 | -0.2641            | -0.2520           | -0.1007 | -0.1097 | -0.0257 |
| Pt    | -0.2411          | -0.2029 | -0.1496 | -0.1593 | -0.2009            | -0.2246           | -0.0950 | -0.0937 | -0.0137 |
| Re    | -0.1869          | -0.2262 | -0.1293 | -0.1697 | -0.1266            | -0.2165           | -0.0644 | -0.0953 | -0.0146 |
| Ta    | -0.1872          | -0.2532 | -0.1523 | -0.1847 | -0.1265            | -0.1798           | -0.0535 | -0.1040 | -0.0177 |
| Tl    | -0.2599          | -0.2833 | -0.1807 | -0.1538 | -0.2422            | -0.1501           | -0.0126 | -0.1338 | -0.0533 |
| W     | -0.1791          | -0.2368 | -0.1335 | -0.1723 | -0.1225            | -0.2253           | -0.0673 | -0.1180 | -0.0201 |

**Table 9:** Zero-point energies in eV for relaxed adsorbates supported on nitrogen-doped graphene at 298.15 K.

| Metal | *CO <sub>2</sub> | *COOH  | *CO    | *CHO   | *CH <sub>2</sub> O | *OCH <sub>3</sub> | *O     | *OH    | *H     |
|-------|------------------|--------|--------|--------|--------------------|-------------------|--------|--------|--------|
| Al    | 0.3093           | 0.6024 | 0.1683 | 0.4395 | 0.7509             | 1.0855            | 0.0552 | 0.3401 | 0.1864 |
| Co    | 0.3140           | 0.6247 | 0.2044 | 0.4787 | 0.7345             | 1.0615            | 0.0606 | 0.3315 | 0.2006 |
| Cr    | 0.3108           | 0.5983 | 0.2007 | 0.4410 | 0.7211             | 1.0824            | 0.0828 | 0.3253 | 0.1625 |
| Cu    | 0.3145           | 0.5943 | 0.1460 | 0.4377 | 0.7196             | 1.0160            | 0.0407 | 0.3104 | 0.1523 |
| Fe    | 0.3117           | 0.6148 | 0.2195 | 0.4634 | 0.7877             | 1.0678            | 0.0703 | 0.3163 | 0.1858 |
| Ga    | 0.3129           | 0.6080 | 0.1561 | 0.4537 | 0.7495             | 1.0826            | 0.0573 | 0.3420 | 0.1897 |
| Ge    | 0.3131           | 0.6206 | 0.1402 | 0.4629 | 0.7281             | 1.1014            | 0.0686 | 0.3538 | 0.2110 |
| Mn    | 0.3120           | 0.6111 | 0.2128 | 0.4601 | 0.8078             | 1.0640            | 0.0793 | 0.3148 | 0.2040 |
| Ni    | 0.3110           | 0.6059 | 0.1485 | 0.4602 | 0.7283             | 1.0467            | 0.0402 | 0.3168 | 0.1620 |
| Sc    | 0.2741           | 0.5910 | 0.1620 | 0.4456 | 0.7019             | 1.0758            | 0.0575 | 0.3166 | 0.1338 |
| Ti    | 0.2920           | 0.6020 | 0.1781 | 0.4525 | 0.7593             | 1.0831            | 0.0731 | 0.2981 | 0.1481 |
| V     | 0.2984           | 0.6055 | 0.1865 | 0.4609 | 0.7756             | 1.0843            | 0.0786 | 0.3171 | 0.1525 |
| Zn    | 0.3143           | 0.5928 | 0.1554 | 0.4357 | 0.7116             | 1.0521            | 0.0394 | 0.3248 | 0.1640 |
| Ag    | 0.3127           | 0.5862 | 0.1862 | 0.4307 | 0.7518             | 1.0419            | 0.0356 | 0.3231 | 0.1308 |
| Cd    | 0.3055           | 0.5910 | 0.1518 | 0.4402 | 0.7312             | 1.0524            | 0.0366 | 0.3189 | 0.1483 |
| In    | 0.3096           | 0.5986 | 0.1414 | 0.4477 | 0.7152             | 1.0724            | 0.0437 | 0.3300 | 0.1609 |
| Mo    | 0.2949           | 0.6111 | 0.1914 | 0.4728 | 0.7806             | 1.0887            | 0.0795 | 0.3253 | 0.1500 |
| Nb    | 0.2946           | 0.5996 | 0.1767 | 0.4601 | 0.7874             | 1.0896            | 0.0742 | 0.3048 | 0.1515 |
| Pd    | 0.3122           | 0.5993 | 0.1444 | 0.4490 | 0.7303             | 1.0342            | 0.0300 | 0.3047 | 0.1558 |
| Rh    | 0.3115           | 0.6288 | 0.1969 | 0.4826 | 0.7388             | 1.0867            | 0.0602 | 0.3453 | 0.2025 |
| Ru    | 0.3152           | 0.6122 | 0.2180 | 0.4569 | 0.7918             | 1.0947            | 0.0698 | 0.3483 | 0.1873 |
| Sb    | 0.3115           | 0.5874 | 0.1397 | 0.4349 | 0.7184             | 1.0467            | 0.0646 | 0.3351 | 0.1752 |
| Sn    | 0.3112           | 0.6002 | 0.1400 | 0.4367 | 0.7239             | 1.0836            | 0.0568 | 0.3359 | 0.1710 |
| Tc    | 0.3053           | 0.6110 | 0.2105 | 0.4251 | 0.7934             | 1.0870            | 0.0807 | 0.3496 | 0.1741 |
| Y     | 0.2709           | 0.5835 | 0.1580 | 0.4307 | 0.7059             | 1.0703            | 0.0500 | 0.3200 | 0.1226 |
| Zr    | 0.2719           | 0.5963 | 0.1723 | 0.4430 | 0.7327             | 1.0846            | 0.0669 | 0.3176 | 0.1472 |
| Au    | 0.3137           | 0.5480 | 0.1442 | 0.3577 | 0.7209             | 0.9864            | 0.0174 | 0.2881 | 0.1027 |
| Bi    | 0.3111           | 0.5778 | 0.1378 | 0.4190 | 0.7133             | 1.0475            | 0.0550 | 0.3241 | 0.1533 |
| Hf    | 0.2699           | 0.6002 | 0.1714 | 0.4452 | 0.7372             | 1.0904            | 0.0687 | 0.3242 | 0.1569 |
| Hg    | 0.3113           | 0.6082 | 0.1397 | 0.4569 | 0.7258             | 1.0720            | 0.0412 | 0.3304 | 0.1625 |
| Ir    | 0.3132           | 0.6309 | 0.2106 | 0.4856 | 0.7390             | 1.0888            | 0.0626 | 0.3456 | 0.2022 |
| Os    | 0.3156           | 0.6148 | 0.2179 | 0.4592 | 0.7971             | 1.0984            | 0.0768 | 0.3463 | 0.1812 |
| Pb    | 0.3116           | 0.5352 | 0.1385 | 0.3677 | 0.7147             | 1.0532            | 0.0474 | 0.3172 | 0.1302 |
| Pt    | 0.3139           | 0.6134 | 0.1439 | 0.4671 | 0.7359             | 1.0406            | 0.0482 | 0.3114 | 0.1736 |
| Re    | 0.3035           | 0.6045 | 0.2095 | 0.4566 | 0.7995             | 1.0957            | 0.0846 | 0.3493 | 0.1818 |
| Ta    | 0.2751           | 0.6019 | 0.1869 | 0.4630 | 0.7724             | 1.0937            | 0.0751 | 0.3060 | 0.1669 |
| Tl    | 0.3131           | 0.5988 | 0.1402 | 0.4439 | 0.7193             | 1.0169            | 0.0573 | 0.3244 | 0.1506 |
| W     | 0.2889           | 0.6117 | 0.1927 | 0.4777 | 0.7852             | 1.0941            | 0.0810 | 0.3273 | 0.1832 |

**Table 10:** Entropy corrections ( $-T \cdot S$ ) in eV for relaxed adsorbates supported on nitrogen-doped graphene at 298.15 K.

| Metal | *CO <sub>2</sub> | *COOH   | *CO     | *CHO    | *CH <sub>2</sub> O | *OCH <sub>3</sub> | *O      | *OH     | *H      |
|-------|------------------|---------|---------|---------|--------------------|-------------------|---------|---------|---------|
| Al    | -0.1981          | -0.2564 | -0.1979 | -0.1424 | -0.2051            | -0.1839           | -0.0794 | -0.1211 | -0.0113 |
| Co    | -0.2358          | -0.2213 | -0.1589 | -0.1584 | -0.2025            | -0.2223           | -0.0701 | -0.0817 | -0.0123 |
| Cr    | -0.2593          | -0.1973 | -0.1568 | -0.2037 | -0.2291            | -0.1679           | -0.0569 | -0.0875 | -0.0228 |
| Cu    | -0.2457          | -0.2636 | -0.2132 | -0.2040 | -0.1267            | -0.2137           | -0.1188 | -0.1711 | -0.0226 |
| Fe    | -0.1936          | -0.2385 | -0.1356 | -0.1758 | -0.1985            | -0.1910           | -0.0647 | -0.0892 | -0.0172 |
| Ga    | -0.2565          | -0.1937 | -0.1600 | -0.2025 | -0.2139            | -0.1766           | -0.0727 | -0.1269 | -0.0126 |
| Ge    | -0.2469          | -0.2413 | -0.1027 | -0.1937 | -0.1687            | -0.1551           | -0.0690 | -0.1091 | -0.0080 |
| Mn    | -0.1929          | -0.2402 | -0.1409 | -0.1736 | -0.1949            | -0.2011           | -0.0563 | -0.1631 | -0.0160 |
| Ni    | -0.1239          | -0.2440 | -0.2027 | -0.1785 | -0.2318            | -0.1792           | -0.0929 | -0.1520 | -0.0170 |
| Sc    | -0.1938          | -0.2647 | -0.1418 | -0.2024 | -0.2291            | -0.2139           | -0.0790 | -0.1516 | -0.0293 |
| Ti    | -0.2093          | -0.2423 | -0.1798 | -0.1640 | -0.1765            | -0.1433           | -0.0692 | -0.1462 | -0.0262 |
| V     | -0.2112          | -0.2373 | -0.1734 | -0.1578 | -0.1642            | -0.0767           | -0.0621 | -0.1522 | -0.0273 |
| Zn    | -0.2567          | -0.2001 | -0.2491 | -0.2135 | -0.1581            | -0.1396           | -0.1091 | -0.1668 | -0.0211 |
| Ag    | -0.2600          | -0.2035 | -0.1802 | -0.2231 | -0.1723            | -0.1531           | -0.1293 | -0.1487 | -0.0596 |
| Cd    | -0.2615          | -0.2042 | -0.1827 | -0.2185 | -0.2727            | -0.2094           | -0.1192 | -0.1850 | -0.0320 |
| In    | -0.1886          | -0.2021 | -0.1593 | -0.2221 | -0.1949            | -0.2061           | -0.1158 | -0.1603 | -0.0238 |
| Mo    | -0.2049          | -0.2075 | -0.1659 | -0.1327 | -0.1650            | -0.2041           | -0.0605 | -0.1258 | -0.0437 |
| Nb    | -0.2072          | -0.2300 | -0.1145 | -0.1439 | -0.1403            | -0.2086           | -0.0684 | -0.1863 | -0.0251 |
| Pd    | -0.1843          | -0.2620 | -0.1476 | -0.2000 | -0.1645            | -0.2057           | -0.1131 | -0.1688 | -0.0223 |
| Rh    | -0.2986          | -0.2220 | -0.1669 | -0.1667 | -0.1953            | -0.1423           | -0.0693 | -0.1229 | -0.0149 |
| Ru    | -0.1998          | -0.1779 | -0.1375 | -0.1827 | -0.1455            | -0.2017           | -0.0653 | -0.1103 | -0.0202 |
| Sb    | -0.1938          | -0.1984 | -0.1084 | -0.2024 | -0.1970            | -0.0996           | -0.0780 | -0.1183 | -0.0109 |
| Sn    | -0.1931          | -0.2040 | -0.1708 | -0.2214 | -0.1801            | -0.2006           | -0.0911 | -0.1474 | -0.0185 |
| Tc    | -0.2038          | -0.2277 | -0.1483 | -0.1505 | -0.1515            | -0.1810           | -0.0569 | -0.1008 | -0.0234 |
| Y     | -0.1994          | -0.2704 | -0.1507 | -0.1605 | -0.2919            | -0.2162           | -0.0981 | -0.1548 | -0.0344 |
| Zr    | -0.2233          | -0.2608 | -0.1845 | -0.1849 | -0.2016            | -0.2144           | -0.0776 | -0.1500 | -0.0242 |
| Au    | -0.1872          | -0.1923 | -0.2237 | -0.1738 | -0.1907            | -0.2639           | -0.1497 | -0.1276 | -0.0301 |
| Bi    | -0.1978          | -0.2736 | -0.1731 | -0.1581 | -0.2369            | -0.0995           | -0.0937 | -0.1386 | -0.0171 |
| Hf    | -0.2235          | -0.2562 | -0.1836 | -0.1839 | -0.1987            | -0.2082           | -0.0759 | -0.1417 | -0.0211 |
| Hg    | -0.1907          | -0.1929 | -0.1053 | -0.0955 | -0.1765            | -0.2558           | -0.1157 | -0.1694 | -0.0357 |
| Ir    | -0.2967          | -0.2218 | -0.1486 | -0.1659 | -0.2038            | -0.2134           | -0.0723 | -0.1326 | -0.0200 |
| Os    | -0.1948          | -0.2462 | -0.1422 | -0.1945 | -0.1448            | -0.2046           | -0.0598 | -0.1205 | -0.0322 |
| Pb    | -0.1979          | -0.2320 | -0.1108 | -0.2414 | -0.1437            | -0.2639           | -0.1126 | -0.1465 | -0.0238 |
| Pt    | -0.1770          | -0.2479 | -0.1481 | -0.1915 | -0.2820            | -0.2030           | -0.0791 | -0.1666 | -0.0228 |
| Re    | -0.1958          | -0.2535 | -0.1508 | -0.1867 | -0.1462            | -0.2483           | -0.0561 | -0.1018 | -0.0249 |
| Ta    | -0.2078          | -0.2197 | -0.1675 | -0.1373 | -0.1542            | -0.2084           | -0.0698 | -0.1979 | -0.0195 |
| Tl    | -0.2573          | -0.2043 | -0.1719 | -0.2270 | -0.1288            | -0.2326           | -0.0688 | -0.1876 | -0.0332 |
| W     | -0.2007          | -0.2007 | -0.1642 | -0.1194 | -0.1565            | -0.2032           | -0.0607 | -0.1275 | -0.0202 |

**Table 11:** Zero-point energies in eV for relaxed adsorbates supported on graphene with dual-vacancy at 298.15 K.

| Metal | *CO <sub>2</sub> | *COOH  | *CO    | *CHO   | *CH <sub>2</sub> O | *OCH <sub>3</sub> | *O     | *OH    | *H     |
|-------|------------------|--------|--------|--------|--------------------|-------------------|--------|--------|--------|
| Al    | 0.3140           | 0.5988 | 0.1888 | 0.4386 | 0.7218             | 1.0769            | 0.0472 | 0.3366 | 0.1729 |
| Co    | 0.3080           | 0.6179 | 0.2173 | 0.4899 | 0.7928             | 1.0616            | 0.0573 | 0.3458 | 0.1726 |
| Cr    | 0.3318           | 0.5862 | 0.1957 | 0.4277 | 0.8826             | 1.0733            | 0.0792 | 0.3294 | 0.1560 |
| Cu    | 0.3140           | 0.6023 | 0.1769 | 0.5229 | 0.7172             | 1.0131            | 0.0323 | 0.3298 | 0.2857 |
| Fe    | 0.3177           | 0.6110 | 0.2083 | 0.5038 | 0.7828             | 1.0802            | 0.0776 | 0.3515 | 0.1987 |
| Ga    | 0.3126           | 0.6056 | 0.1492 | 0.4494 | 0.7196             | 1.0788            | 0.0446 | 0.3370 | 0.1810 |
| Ge    | 0.3138           | 0.6171 | 0.1471 | 0.4657 | 0.7196             | 1.0935            | 0.0540 | 0.3514 | 0.2028 |
| Mn    | 0.3167           | 0.6141 | 0.2050 | 0.4803 | 0.8009             | 1.0809            | 0.0756 | 0.3411 | 0.1594 |
| Ni    | 0.3134           | 0.6251 | 0.1942 | 0.4571 | 0.7230             | 1.0738            | 0.0496 | 0.3521 | 0.2877 |
| Sc    | 0.3229           | 0.5874 | 0.1668 | 0.4367 | 0.7358             | 1.0673            | 0.0533 | 0.3129 | 0.1288 |
| Ti    | 0.3240           | 0.6049 | 0.1749 | 0.4562 | 0.7202             | 1.0694            | 0.0670 | 0.3143 | 0.1559 |
| V     | 0.2937           | 0.6042 | 0.1933 | 0.4593 | 0.8437             | 1.0745            | 0.0753 | 0.3319 | 0.1687 |
| Zn    | 0.3123           | 0.5888 | 0.1564 | 0.4317 | 0.7111             | 1.0363            | 0.0324 | 0.3157 | 0.1363 |
| Ag    | 0.3172           | 0.5984 | 0.1399 | 0.4312 | 0.7199             | 1.0034            | 0.0258 | 0.3141 | 0.2932 |
| Cd    | 0.3120           | 0.5796 | 0.1487 | 0.4176 | 0.7144             | 1.0198            | 0.0278 | 0.3084 | 0.1280 |
| In    | 0.3134           | 0.5973 | 0.1458 | 0.4443 | 0.7170             | 1.0666            | 0.0418 | 0.3236 | 0.1610 |
| Mo    | 0.3052           | 0.5923 | 0.1942 | 0.4668 | 0.7718             | 1.0844            | 0.0745 | 0.3350 | 0.1755 |
| Nb    | 0.2952           | 0.6071 | 0.1862 | 0.4605 | 0.7505             | 1.0749            | 0.0721 | 0.3274 | 0.1717 |
| Pd    | 0.3151           | 0.6191 | 0.1872 | 0.4609 | 0.7187             | 1.0525            | 0.0397 | 0.3371 | 0.2963 |
| Rh    | 0.2870           | 0.6096 | 0.2076 | 0.4297 | 0.7850             | 1.0628            | 0.0604 | 0.3590 | 0.2678 |
| Ru    | 0.3000           | 0.6040 | 0.2097 | 0.4645 | 0.7812             | 1.0710            | 0.0790 | 0.3319 | 0.1758 |
| Sb    | 0.3122           | 0.6143 | 0.1436 | 0.4604 | 0.7258             | 1.0938            | 0.0608 | 0.3481 | 0.1912 |
| Sn    | 0.3132           | 0.6077 | 0.1493 | 0.4550 | 0.7182             | 1.0807            | 0.0517 | 0.3338 | 0.1770 |
| Tc    | 0.3051           | 0.6025 | 0.2086 | 0.4585 | 0.7777             | 1.0690            | 0.0799 | 0.3285 | 0.2898 |
| Y     | 0.3219           | 0.5847 | 0.1632 | 0.4299 | 0.7245             | 1.0646            | 0.0557 | 0.3182 | 0.1164 |
| Zr    | 0.3213           | 0.5966 | 0.1669 | 0.4527 | 0.7200             | 1.0620            | 0.0568 | 0.3151 | 0.1510 |
| Au    | 0.3172           | 0.5927 | 0.1448 | 0.4677 | 0.7261             | 0.9827            | 0.0291 | 0.2904 | 0.2898 |
| Bi    | 0.3108           | 0.6062 | 0.1406 | 0.4521 | 0.7185             | 1.0309            | 0.0528 | 0.3289 | 0.1696 |
| Hf    | 0.3241           | 0.5981 | 0.1713 | 0.4543 | 0.7236             | 1.0751            | 0.0586 | 0.3142 | 0.1550 |
| Hg    | 0.3130           | 0.5763 | 0.1421 | 0.3959 | 0.7161             | 0.9951            | 0.0177 | 0.2917 | 0.1258 |
| Ir    | 0.3054           | 0.6129 | 0.2147 | 0.4199 | 0.7939             | 1.0680            | 0.0712 | 0.3603 | 0.1863 |
| Os    | 0.3033           | 0.6068 | 0.2155 | 0.4695 | 0.7674             | 1.0779            | 0.0823 | 0.3330 | 0.1401 |
| Pb    | 0.3118           | 0.6044 | 0.1424 | 0.4518 | 0.7199             | 1.0740            | 0.0502 | 0.3250 | 0.1623 |
| Pt    | 0.3124           | 0.6242 | 0.1985 | 0.4680 | 0.7263             | 1.0514            | 0.0477 | 0.3406 | 0.2725 |
| Re    | 0.3033           | 0.6034 | 0.2108 | 0.4601 | 0.7864             | 1.0807            | 0.0780 | 0.3317 | 0.2191 |
| Ta    | 0.2938           | 0.6078 | 0.1872 | 0.4601 | 0.7707             | 1.0754            | 0.0732 | 0.3243 | 0.1783 |
| Tl    | 0.3091           | 0.5948 | 0.1424 | 0.4406 | 0.7123             | 1.0568            | 0.0342 | 0.3173 | 0.1569 |
| W     | 0.3066           | 0.6062 | 0.1912 | 0.3984 | 0.8010             | 1.0777            | 0.0787 | 0.3333 | 0.1815 |

**Table 12:** Entropy corrections ( $-T \cdot S$ ) in eV for relaxed adsorbates supported on graphene with dual-vacancy at 298.15 K.

| Metal | *CO <sub>2</sub> | *COOH   | *CO     | *CHO    | *CH <sub>2</sub> O | *OCH <sub>3</sub> | *O      | *OH     | *H      |
|-------|------------------|---------|---------|---------|--------------------|-------------------|---------|---------|---------|
| Al    | -0.2501          | -0.2622 | -0.1799 | -0.2106 | -0.3101            | -0.2050           | -0.0980 | -0.1303 | -0.0165 |
| Co    | -0.1962          | -0.2352 | -0.1529 | -0.1520 | -0.1515            | -0.0821           | -0.0427 | -0.1165 | -0.0376 |
| Cr    | -0.2178          | -0.2281 | -0.1603 | -0.2297 | -0.0849            | -0.0961           | -0.0623 | -0.1200 | -0.0250 |
| Cu    | -0.1907          | -0.2528 | -0.1276 | -0.1239 | -0.1974            | -0.2279           | -0.0711 | -0.1357 | -0.0044 |
| Fe    | -0.2328          | -0.2473 | -0.0848 | -0.1300 | -0.1611            | -0.1248           | -0.0629 | -0.0996 | -0.0292 |
| Ga    | -0.1842          | -0.2577 | -0.2003 | -0.2061 | -0.1912            | -0.2332           | -0.0956 | -0.1395 | -0.0145 |
| Ge    | -0.2491          | -0.1850 | -0.1442 | -0.1942 | -0.1974            | -0.1522           | -0.0886 | -0.1119 | -0.0101 |
| Mn    | -0.2384          | -0.2493 | -0.1559 | -0.1654 | -0.1993            | -0.1304           | -0.0599 | -0.1090 | -0.0275 |
| Ni    | -0.2497          | -0.2279 | -0.1683 | -0.1836 | -0.2402            | -0.2254           | -0.1159 | -0.1027 | -0.0021 |
| Sc    | -0.2252          | -0.1996 | -0.1449 | -0.2023 | -0.2586            | -0.2259           | -0.1128 | -0.1670 | -0.0322 |
| Ti    | -0.2137          | -0.2297 | -0.1918 | -0.1703 | -0.2184            | -0.2455           | -0.0745 | -0.1598 | -0.0186 |
| V     | -0.2319          | -0.2441 | -0.1693 | -0.1713 | -0.0933            | -0.0952           | -0.0673 | -0.1171 | -0.0164 |
| Zn    | -0.1971          | -0.2071 | -0.1798 | -0.2147 | -0.1577            | -0.2074           | -0.1203 | -0.1637 | -0.0293 |
| Ag    | -0.2465          | -0.2579 | -0.1696 | -0.1739 | -0.1930            | -0.2345           | -0.0117 | -0.1000 | -0.0038 |
| Cd    | -0.1963          | -0.2145 | -0.2124 | -0.2350 | -0.2120            | -0.2341           | -0.1338 | -0.1754 | -0.0354 |
| In    | -0.2567          | -0.2013 | -0.1464 | -0.1558 | -0.2003            | -0.1995           | -0.1043 | -0.1065 | -0.0228 |
| Mo    | -0.2106          | -0.2086 | -0.1656 | -0.1746 | -0.1628            | -0.2038           | -0.0668 | -0.1129 | -0.0144 |
| Nb    | -0.2307          | -0.2394 | -0.1752 | -0.1660 | -0.1679            | -0.1668           | -0.0688 | -0.1288 | -0.0138 |
| Pd    | -0.3134          | -0.2362 | -0.1751 | -0.1707 | -0.1779            | -0.0819           | -0.0677 | -0.1222 | -0.0039 |
| Rh    | -0.2128          | -0.2468 | -0.1542 | -0.1986 | -0.1595            | -0.0720           | -0.0949 | -0.0933 | -0.0021 |
| Ru    | -0.2275          | -0.2613 | -0.1426 | -0.1760 | -0.1693            | -0.0786           | -0.0577 | -0.1332 | -0.0175 |
| Sb    | -0.1891          | -0.1940 | -0.2266 | -0.1526 | -0.2398            | -0.2284           | -0.0788 | -0.1157 | -0.0132 |
| Sn    | -0.2529          | -0.1968 | -0.2028 | -0.2165 | -0.1898            | -0.1845           | -0.0886 | -0.1568 | -0.0170 |
| Tc    | -0.2158          | -0.2603 | -0.1459 | -0.1731 | -0.1578            | -0.0973           | -0.0574 | -0.1222 | -0.0647 |
| Y     | -0.2962          | -0.2657 | -0.1496 | -0.2075 | -0.2598            | -0.2231           | -0.1096 | -0.1662 | -0.0384 |
| Zr    | -0.1548          | -0.2417 | -0.1360 | -0.1770 | -0.2745            | -0.2015           | -0.0919 | -0.1551 | -0.0185 |
| Au    | -0.2413          | -0.2076 | -0.2877 | -0.1138 | -0.2448            | -0.2634           | -0.0095 | -0.1447 | -0.0041 |
| Bi    | -0.1970          | -0.2006 | -0.2337 | -0.1648 | -0.1315            | -0.2394           | -0.0927 | -0.1399 | -0.0183 |
| Hf    | -0.2728          | -0.2398 | -0.1920 | -0.1784 | -0.2711            | -0.3070           | -0.0834 | -0.1620 | -0.0186 |
| Hg    | -0.1953          | -0.2848 | -0.1710 | -0.1987 | -0.2065            | -0.1285           | -0.1491 | -0.1469 | -0.0370 |
| Ir    | -0.2144          | -0.1796 | -0.1476 | -0.1609 | -0.1474            | -0.0751           | -0.0699 | -0.0953 | -0.0258 |
| Os    | -0.2163          | -0.2409 | -0.1420 | -0.1599 | -0.1776            | -0.0816           | -0.0565 | -0.1218 | -0.0724 |
| Pb    | -0.1964          | -0.2001 | -0.2339 | -0.1644 | -0.1894            | -0.1890           | -0.0888 | -0.1086 | -0.0227 |
| Pt    | -0.1825          | -0.2340 | -0.1657 | -0.1759 | -0.2307            | -0.1921           | -0.0612 | -0.1216 | -0.0022 |
| Re    | -0.2117          | -0.2566 | -0.1472 | -0.1661 | -0.1530            | -0.1479           | -0.0619 | -0.1223 | -0.0261 |
| Ta    | -0.2269          | -0.2462 | -0.1744 | -0.1715 | -0.1696            | -0.1787           | -0.0694 | -0.1428 | -0.0135 |
| Tl    | -0.1981          | -0.2044 | -0.2304 | -0.2198 | -0.2048            | -0.2595           | -0.1146 | -0.1764 | -0.0256 |
| W     | -0.2006          | -0.2433 | -0.1670 | -0.1453 | -0.1295            | -0.1025           | -0.0613 | -0.1185 | -0.0157 |



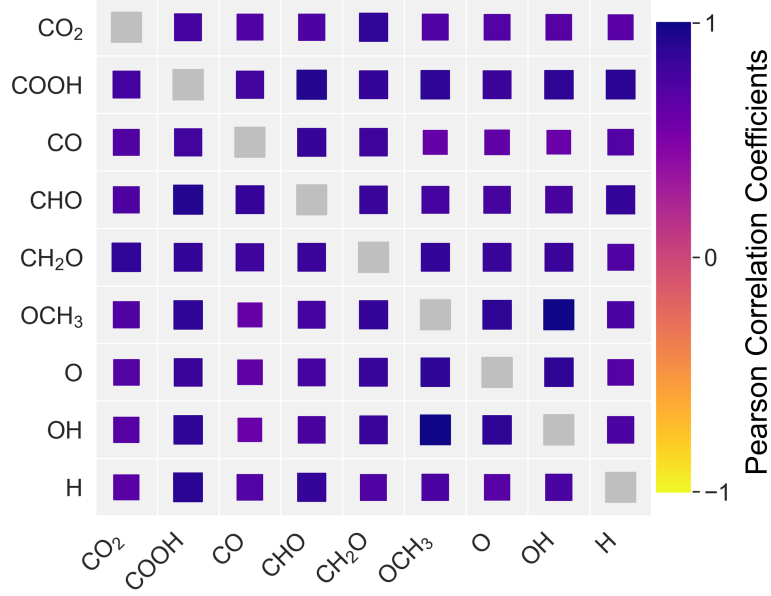


Figure 15: Pearson correlation map for adsorption energies of various adsorbates for all six substrates.

## 1.5 The scaling relation scheme and the hybrid scaling relation

In line with the scaling relation framework proposed by Abild-Pedersen et al. [5], and later applied to the CO<sub>2</sub> reduction to methane process by Peterson & Nørskov [6], adsorbates in the CO<sub>2</sub>RR process can be categorized into C-centered or O-centered species. In each group, a representative species is designated as the “descriptor”, allowing approximating the adsorption energies of other species within the same group using scaling relations. The following equations capture this relationship:

$$G_{\text{ads}}X = a_X \cdot G_{\text{ads}}\text{CO} + c_X \quad (\text{where } X = \text{COOH}, \text{CHO}, \text{CH}_2\text{O}) \quad (1)$$

$$G_{\text{ads}}Y = a_Y \cdot G_{\text{ads}}\text{OH} + c_Y \quad (\text{where } Y = \text{O}, \text{OCH}_3) \quad (2)$$

Here, CO and OH serve as descriptors for C-centered ( $X$ ) and O-centered ( $Y$ ) groups, respectively, and  $a_X, c_X, a_Y, c_Y$  are species-specific scaling parameters. The scaling relations simplify the adsorption energy landscape to a two-dimensional space, thereby enabling visualization.

However, it’s crucial to note that in the CO<sub>2</sub>RR process, most species are not solely C- or O-centered. Thus, it becomes advantageous to incorporate both

types of species in the scaling relation. Building upon the hybrid scaling relation introduced in the main text, the free energy change ( $\Delta G$ ) for any given reaction step can be readily calculated using Equation (14) from the main text. Consider the generic reaction step:



The free energy change  $\Delta G$  is defined as:

$$\Delta G = G(*B) - G(*A) - G(H^+) - G(e^-) \quad (4)$$

Focusing on any reaction step delineated by Equation (1) to Equation (8) in the main text, Equation (15) shows that limiting potential  $U_L$  at a specific external potential  $U$  is fully determined by the adsorption energies of two descriptors. For instance, for the reaction described by Equation (3):

$$\Delta G = G(*COOH) - G(*) - G(CO_2) - G(H^+) - G(e^-) \quad (5)$$

$$= G(COOH) + G(*) + G_{ads}COOH - G(*) - G(CO_2) - G(H^+) - G(e^-) \quad (6)$$

$$= G_{ads}COOH + G(COOH) - G(CO_2) - G(H^+) - G(e^-) \quad (7)$$

From adsorption energy linear relations, we have:

$$G_{ads}COOH = a_{COOH} \cdot G_{ads}CO + b_{COOH} \cdot G_{ads}OH + c_{COOH} \quad (8)$$

which then reduces Supplementary Equation (7) to:

$$\Delta G = (a_{COOH} \cdot G_{ads}CO + b_{COOH} \cdot G_{ads}OH) + [c_{COOH} + G(COOH) - G(CO_2) - G(H^+) - G(e^-)] \quad (9)$$

In this case, the first term depends solely on the descriptors, while the second term remains constant at a given potential  $U$ .

## 1.6 Volcano plots

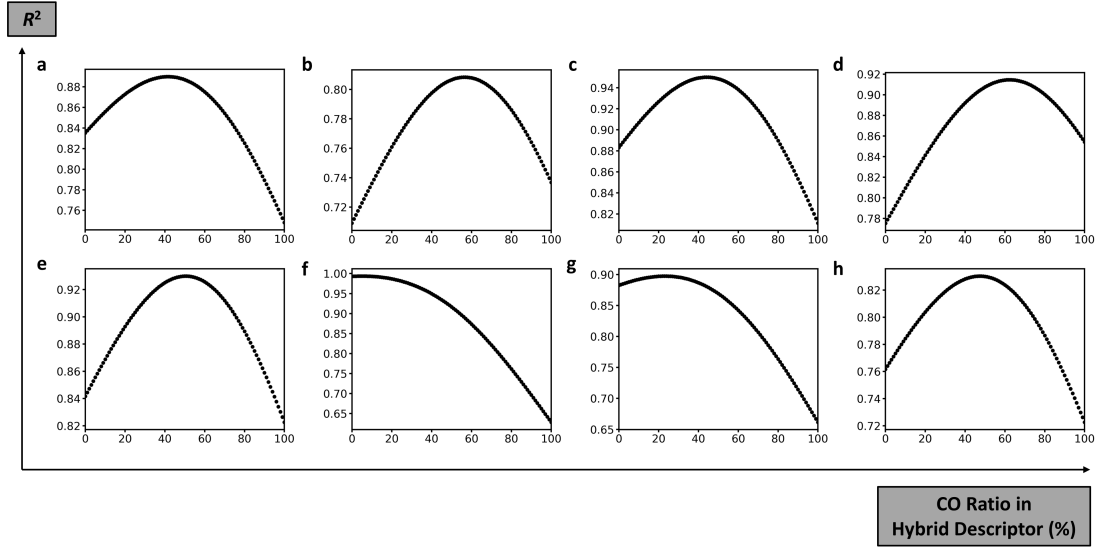
In addition to the selectivity volcano plot featured in the main text, we provide a comprehensive plot capturing all rate-determining steps for CO<sub>2</sub>RR in Figure 17. A separate plot focusing on selectivity over HER is also presented in Figure 18.

## 1.7 Activity periodic tables

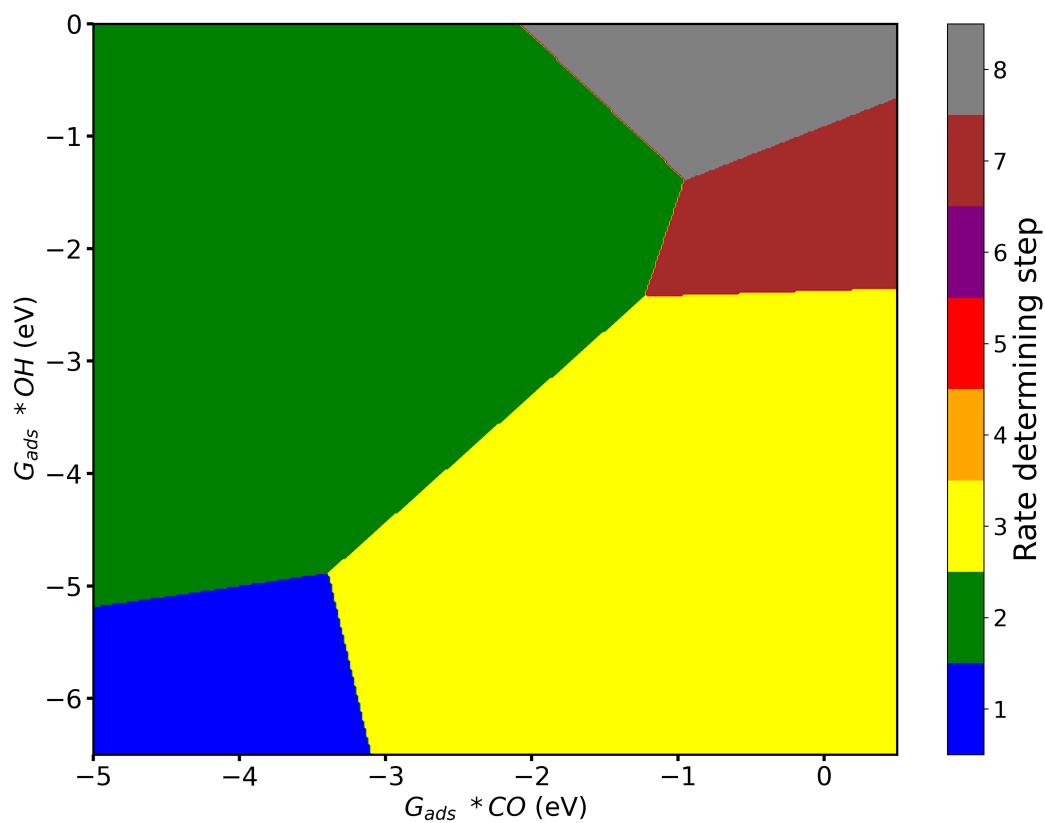
**Table 13:** Adsorption free energy scaling relation parameters determined by the hybrid descriptor method.

| Adsorbate         | $a_Z$  | $b_Z$  | $c_Z$   |
|-------------------|--------|--------|---------|
| COOH              | 0.4032 | 0.5132 | 0.0899  |
| CHO               | 0.5977 | 0.3664 | -2.6763 |
| CH <sub>2</sub> O | 0.5011 | 0.5011 | -0.9806 |
| OCH <sub>3</sub>  | 0.0532 | 1.0114 | 0.8489  |
| O                 | 0.3758 | 1.2580 | 0.0910  |
| H                 | 0.3164 | 0.3568 | -0.6764 |

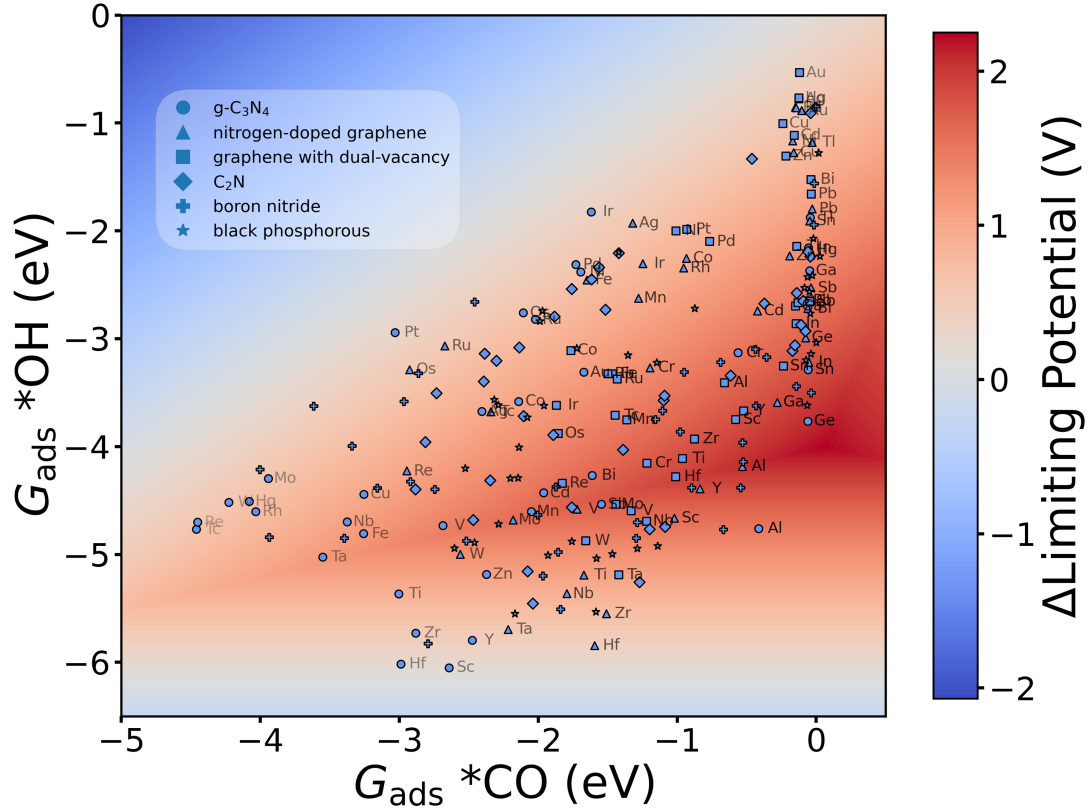
*Note:*  $a_Z$ ,  $b_Z$  and  $c_Z$  correspond to coefficients for  $G_{\text{ads}}\text{CO}$ ,  $G_{\text{ads}}\text{OH}$  and constant terms, where  $Z$  stands for the adsorbed species of interest.



**Figure 16:**  $R^2$  improvement from hybrid-descriptor method. Improvement in the coefficient of determination ( $R^2$ ) with x-axis as CO ratio in CO/OH hybrid descriptors. (a) averaged  $R^2$  and those of (b) CO<sub>2</sub>, (c) COOH, (d) CHO, (e) CH<sub>2</sub>O, (f) OCH<sub>3</sub>, (g) O and (h) H are shown.



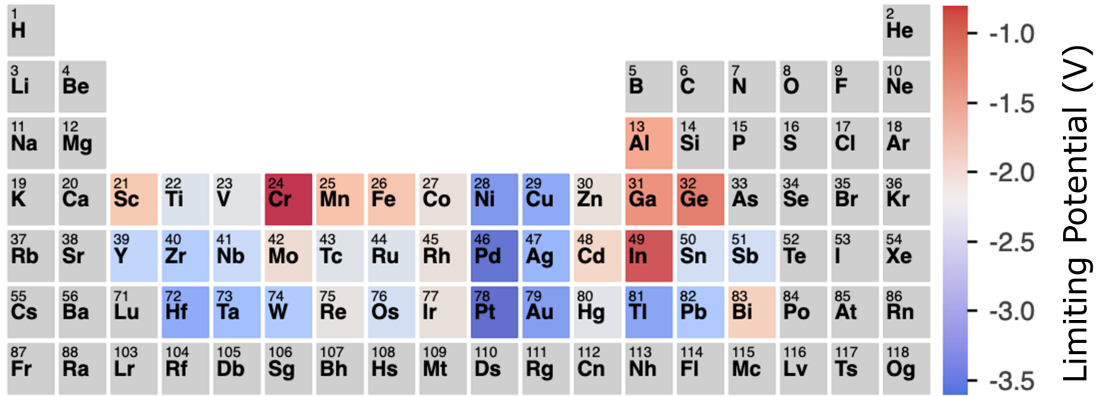
**Figure 17: CO<sub>2</sub>RR rate determining step volcano plot.** This graph illustrates the rate-determining step associated with each region within the CO<sub>2</sub>RR process. Roman numerals denoting the respective RDS for each region.



**Figure 18: CO<sub>2</sub>RR selectivity volcano plot.** Selectivity volcano plot computed from linear scaling relations. In the plot, the circle, triangle, square, diamond, plus, and star symbols correspond to metals supported on g-C<sub>3</sub>N<sub>4</sub>, nitrogen-doped graphene, graphene with dual-vacancy, black phosphorous, BN, and C<sub>2</sub>N, respectively.

**Table 14:** Performance comparison of hybrid descriptor and single descriptor for linear scaling relations.

| Adsorbate         | Hybrid Descriptor $R^2$<br>(Optimal CO Ratio) | Single Descriptor $R^2$<br>Reference Species | $R^2$  |
|-------------------|-----------------------------------------------|----------------------------------------------|--------|
| CO <sub>2</sub>   | 0.8083 (57%)                                  | 0.7368 (CO)                                  | 0.0715 |
| COOH              | 0.9500 (44%)                                  | 0.8113 (CO)                                  | 0.1387 |
| CHO               | 0.9146 (62%)                                  | 0.8541 (CO)                                  | 0.0605 |
| CH <sub>2</sub> O | 0.9299 (50%)                                  | 0.8226 (CO)                                  | 0.1073 |
| OCH <sub>3</sub>  | 0.9936 (5%)                                   | 0.9929 (OH)                                  | 0.0007 |
| O                 | 0.8974 (23%)                                  | 0.8825 (OH)                                  | 0.0149 |
| H                 | 0.8303 (47%)                                  | 0.7617 (OH)                                  | 0.0686 |



**Figure 19:** Limiting potential periodic table for nitrogen-doped graphene.

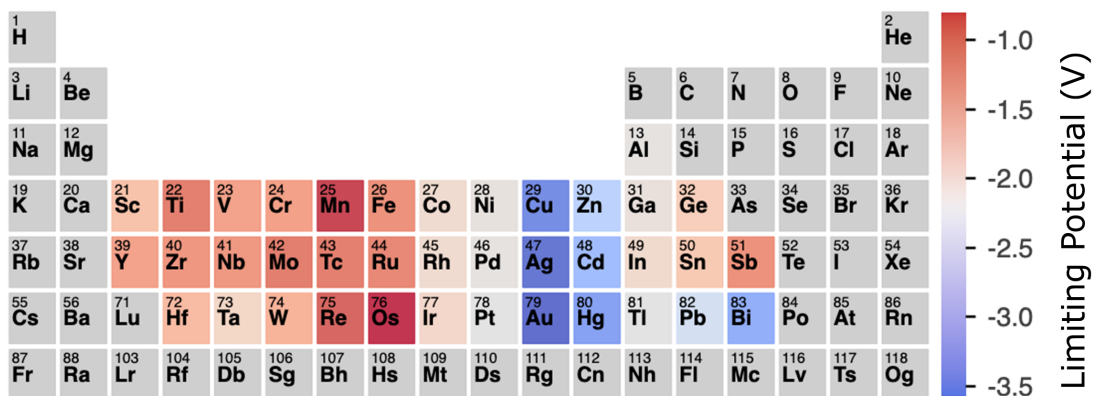


Figure 20: Limiting potential periodic table for graphene with dual-vacancy.

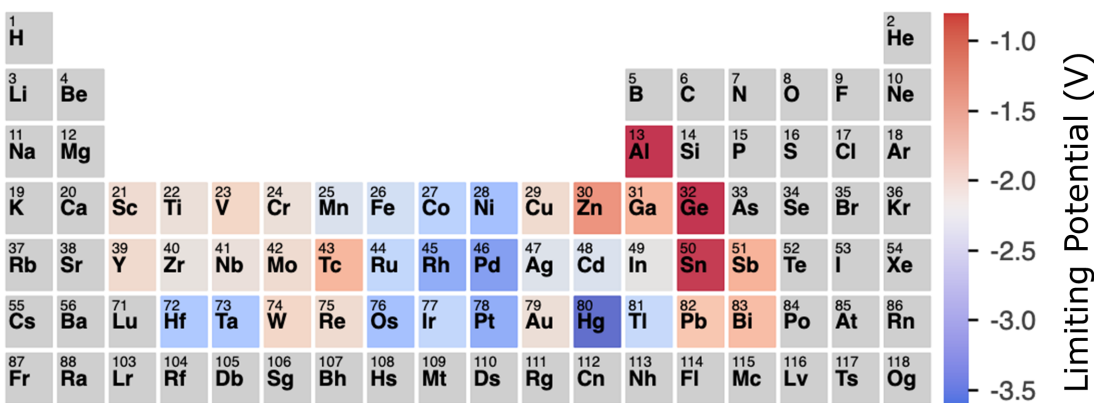


Figure 21: Limiting potential periodic table for black phosphorus.

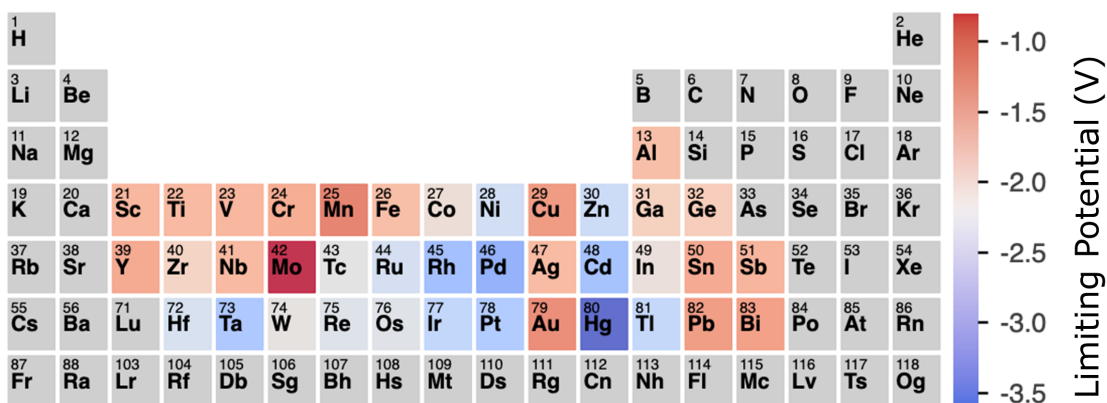


Figure 22: Limiting potential periodic table for boron nitride.

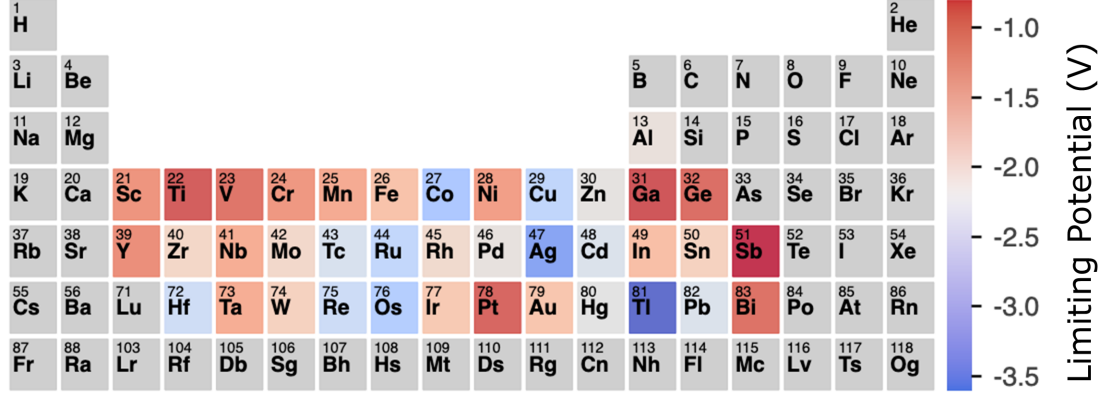


Figure 23: Limiting potential periodic table for monolayer  $C_2N$ .

## 2 Additional details on machine learning method

### 2.1 d-band center and adsorption energy relations

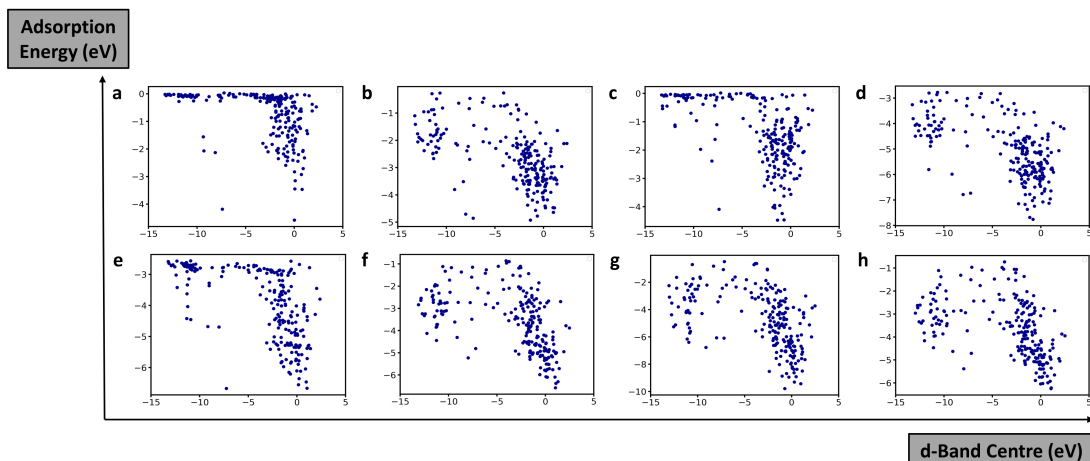
### 2.2 Feature correlation analysis

Figure 25 displays the pairwise relationships between CO adsorption energy and its four most correlated descriptors: d-band center (spin-up)  $\delta\epsilon_d \uparrow$ , lattice parameter  $\gamma$ , vacuum level  $E_{vac}$ , and electronegativity on the Allen scale  $\chi_{Allen}$ , identified by Kendall rank correlation analysis. section 2.2 lists notations for elementary descriptors and electronic descriptors.

**Table 15:** Notations for elementary descriptors and electronic descriptors.

| Elementary Descriptors              |                                                                | Electronic Descriptors              |                                          |
|-------------------------------------|----------------------------------------------------------------|-------------------------------------|------------------------------------------|
| $R$                                 | atomic radius                                                  | $\alpha, \beta, \gamma$             | lattice parameters                       |
| $E_{ea}$                            | electron affinity                                              | $\Phi_{DFT}$                        | DFT calculated work function             |
| $\Phi$                              | work function                                                  | $E_{vac}$                           | vacuum level                             |
| $\chi_{Allen}, \chi_P, \chi_{RevP}$ | electronegativity in Allen, Pauling and revised Pauling scales | $E_g, E_g \uparrow, E_g \downarrow$ | average, spin-up and spin-down band gaps |
| $A_r$                               | relative atomic mass                                           | $e_d$                               | number of d-electrons                    |
| $E_i$                               | ionization energy                                              | $\delta\epsilon_d \uparrow$         | d-band centre (spin-up)                  |
| $G$                                 | group number                                                   | $W_d$                               | d-band width                             |
| $P$                                 | period number                                                  | $e_{Bader}$                         | Bader charge                             |
| $V$                                 | number of valence electrons                                    |                                     |                                          |





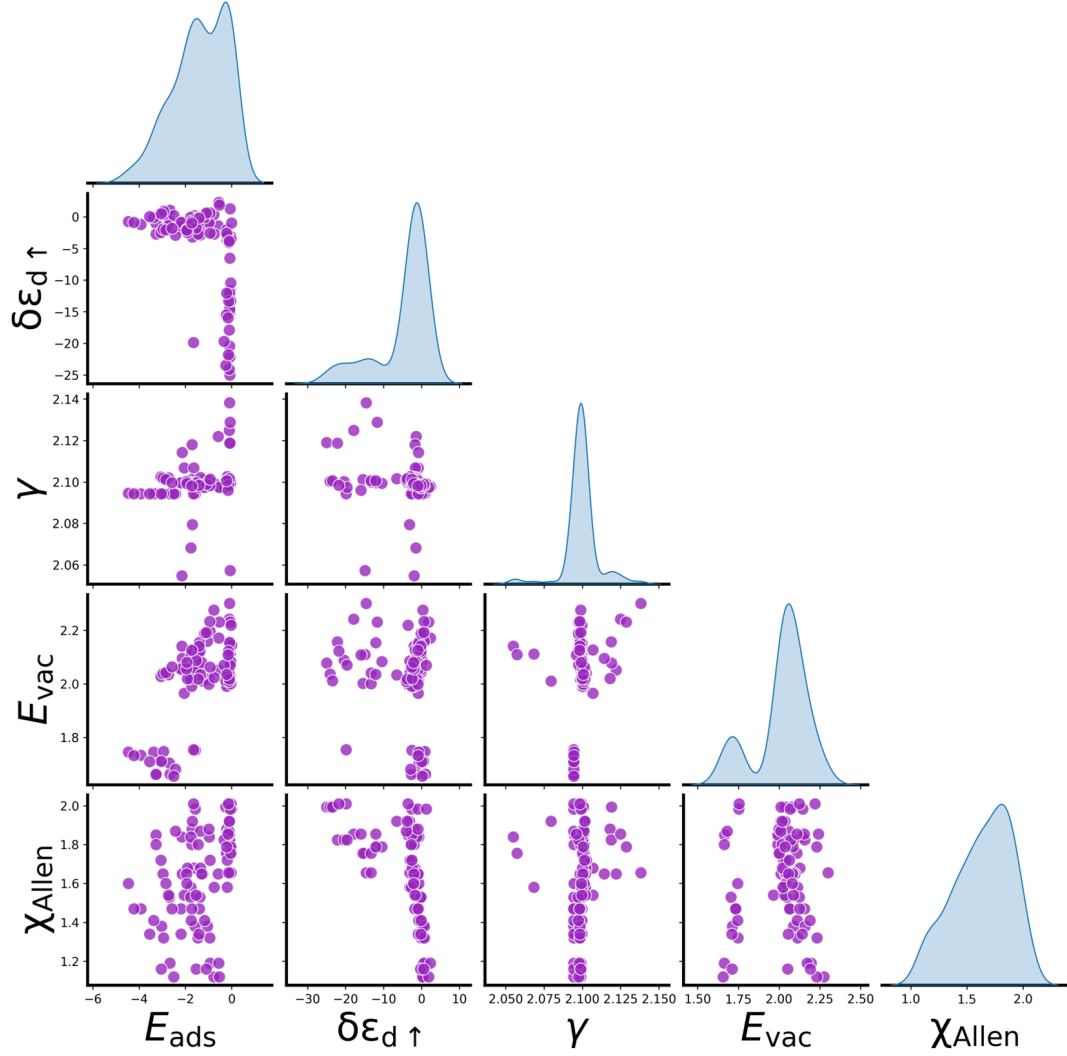
**Figure 24: Scatter plot of d-Band Center and Adsorption Energy Relationships.** This figure delineates the relationship between d-band center and adsorption energies for various species: (a)  $\text{CO}_2$ , (b)  $\text{COOH}$ , (c)  $\text{CO}$ , (d)  $\text{CHO}$ , (e)  $\text{CH}_2\text{O}$ , (f)  $\text{OCH}_3$ , (g)  $\text{O}$ , and (h)  $\text{OH}$ . Data from all six substrates investigated in this study are included.

## 2.3 Input pipeline construction

The format of the 2D array for the eDOS is naturally well-suited for use as neural network inputs. In this study, the array adopts a shape of  $[\text{numSamplings}, \text{numOrbitals}, \text{numChannels}]$ , as illustrated in Figure 26. Here, “numSamplings” represents the number of data points sampled across the energy spectrum, “numOrbitals” denotes the number of DOS orbitals, and “numChannels” indicates the number of atomic channels.

For practical purposes, certain spatial hierarchies are expected to remain invariant, such as the energy range over which the eDOS is sampled. In this work, we maintain a consistent energy range of -14 eV to 6 eV, sampled at a density of  $50 \text{ points} \cdot \text{eV}^{-1}$ . In addition to maintaining a fixed energy range, the following steps are taken to standardize the input eDOS data: 1. For atoms lacking d-electrons, the eDOS is zero-padded. 2. f-orbital eDOS data are excluded. 3. The eDOS of adsorbates is zero-padded along the Channels axis to align with the adsorbate that contains the maximum number of atoms, which, in this study, is the  $\text{*OCH}_3$  species. This padded eDOS is then appended to the metal eDOS.

As a result of these preprocessing steps, the input eDOS arrays consistently have a shape of  $[4000, 9, 6]$ . This corresponds to 4000 sampling points across the energy range, 9 eDOS orbitals, and 6 atomic channels.



**Figure 25: Pairwise Plot of Adsorption Energy and Key Descriptors.** This chart displays the pairwise relationships between CO adsorption energy and its four most correlated descriptors: d-band center (spin-up)  $\delta\epsilon_{\text{d}} \uparrow$ , lattice parameter  $\gamma$ , vacuum level  $E_{\text{vac}}$ , and electronegativity on the Allen scale  $\chi_{\text{Allen}}$ , identified by Kendall rank correlation analysis. Data from all six substrates investigated in this study are included.

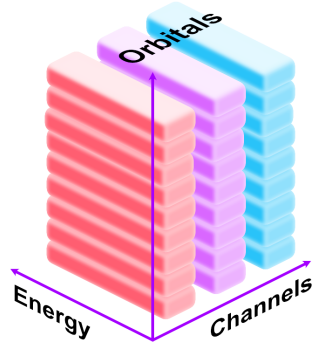


Figure 26: Schematic representation of the input eDOS data structure.

## 2.4 Hyperparameter tuning for CNN

Table 16: The hyperparameter search space.

| Hyperparameter | Search Space                     | Description                                                              |
|----------------|----------------------------------|--------------------------------------------------------------------------|
| learningRate   | $[10^{-4}, 10^{-2}]$ , log scale | Learning rate of Adam [7] optimizer                                      |
| dropoutRate    | $[0, 1]$                         | Dropout rate                                                             |
| root_fc0       | 128, 256, 512                    | Output dimensionality of 1 <sup>st</sup> fully connected layer in root   |
| root_fc1       | 256, 512, 1024, 2048             | Output dimensionality of 2 <sup>nd</sup> fully connected layer in root   |
| root_act       | "tanh", "relu", "sigmoid"        | Activation function of fully connected layers in root                    |
| kernelSize     | $[2, 32]$                        | Width of the convolution window                                          |
| numFilters     | $[8, 64]$                        | Number of filters in the convolution                                     |
| numConvLayers  | $[1, 16]$                        | Number of convolution layers                                             |
| numConvBlocks  | $[1, 16]$                        | Number of convolution blocks                                             |
| br_fc0         | 16, 32, 64, 128, 256, 512        | Output dimensionality of 1 <sup>st</sup> fully connected layer in branch |
| br_fc1         | 8, 16, 32, 64, 128, 256          | Output dimensionality of 2 <sup>nd</sup> fully connected layer in branch |
| br_act         | "tanh", "relu", "sigmoid"        | Activation function of fully connected layers in branch                  |

## 2.5 Occlusion experiment

## 2.6 Shifting experiment

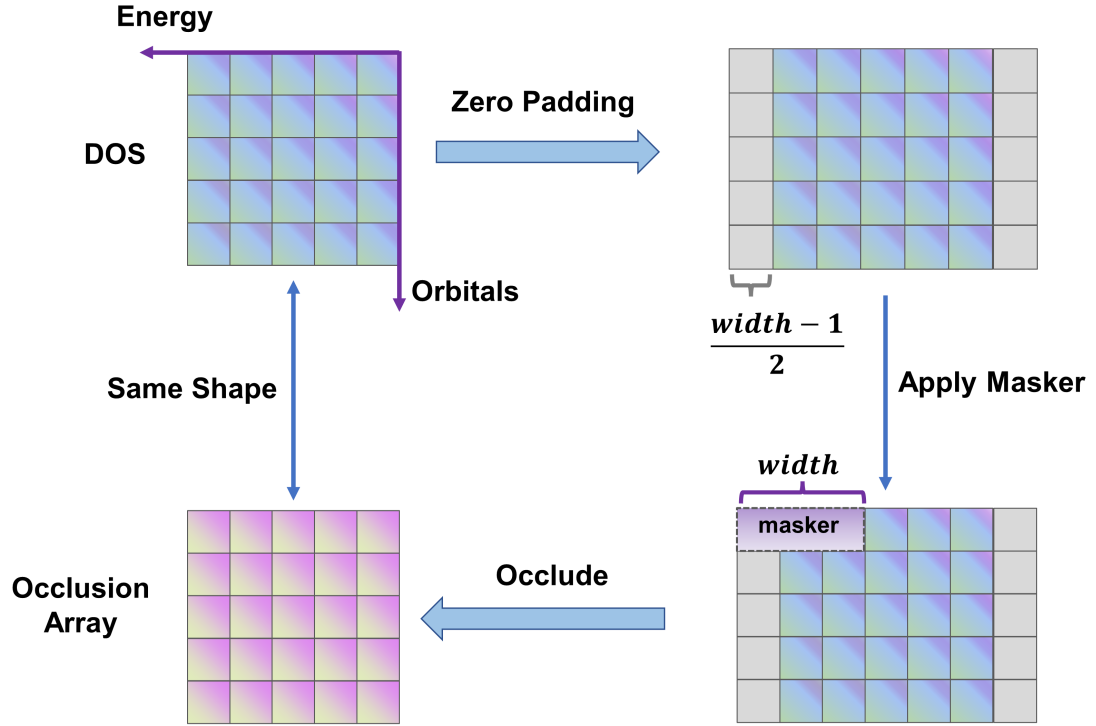
**Table 17:** Optimal hyperparameters identified with Hyperband algorithm.

| Hyperparameter | Optimal Value | Description                                                              |
|----------------|---------------|--------------------------------------------------------------------------|
| learningRate   | $10^{-3}$     | Learning rate of Adam [7] optimizer                                      |
| dropoutRate    | 0.3           | Dropout rate                                                             |
| root_fc0       | 256           | Output dimensionality of 1 <sup>st</sup> fully connected layer in root   |
| root_fc1       | 512           | Output dimensionality of 2 <sup>nd</sup> fully connected layer in root   |
| root_act       | “relu”        | Activation function of fully connected layers in root                    |
| kernelSize     | 10            | Width of convolution window                                              |
| numFilters     | 17            | Number of filters in the convolution                                     |
| numConvLayers  | 3             | Number of convolution layers                                             |
| numConvBlocks  | 1             | Number of convolution blocks                                             |
| br_fc0         | 256           | Output dimensionality of 1 <sup>st</sup> fully connected layer in branch |
| br_fc1         | 128           | Output dimensionality of 2 <sup>nd</sup> fully connected layer in branch |
| br_act         | “tanh”        | Activation function of fully connected layers in branch                  |

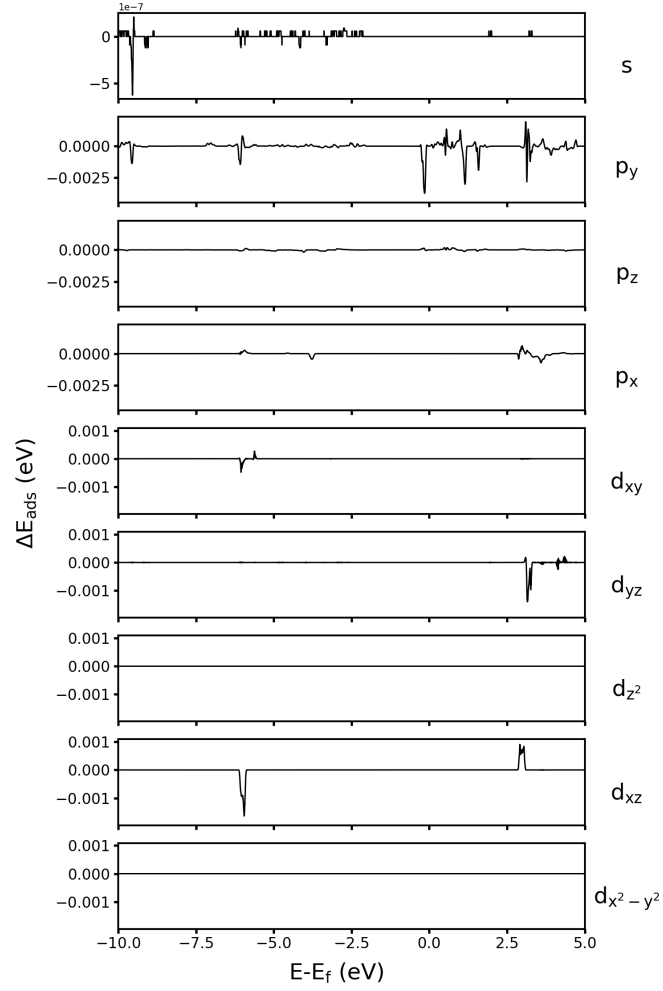
**Table 18:** Prediction mean absolute errors of the CNN model.

| Adsorbate         | Original dataset (eV) | Augmented dataset (eV) |
|-------------------|-----------------------|------------------------|
| CO <sub>2</sub>   | 0.0447                | 0.0336                 |
| COOH              | 0.0436                | 0.0581                 |
| CO                | 0.0402                | 0.0402                 |
| CHO               | 0.0614                | 0.0778                 |
| CH <sub>2</sub> O | 0.0553                | 0.0632                 |
| OCH <sub>3</sub>  | 0.0605                | 0.0636                 |
| O                 | 0.1135                | 0.1081                 |
| OH                | 0.0547                | 0.0602                 |
| H                 | 0.0703                | 0.0646                 |

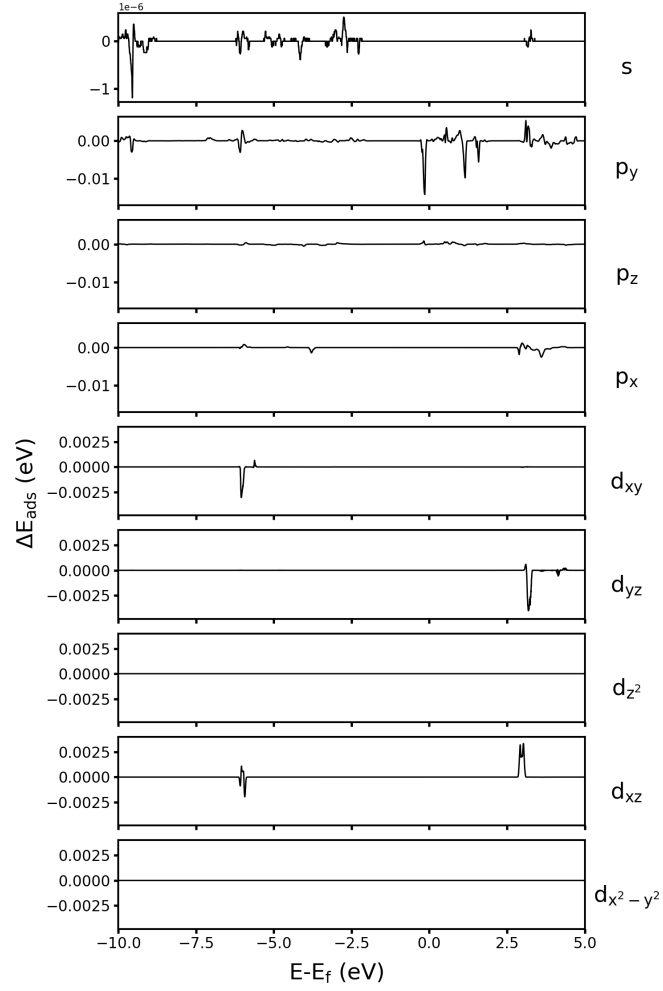
*Note:* The same CNN model, trained on the augmented dataset, was used for all predictions. The term “original dataset” refers to evaluations conducted on the original dataset without any augmented data; it does not imply the use of a different CNN model trained solely on the original dataset. Similarly, “augmented dataset” refers to evaluations on the dataset with both original data and augment data, not a distinct model.



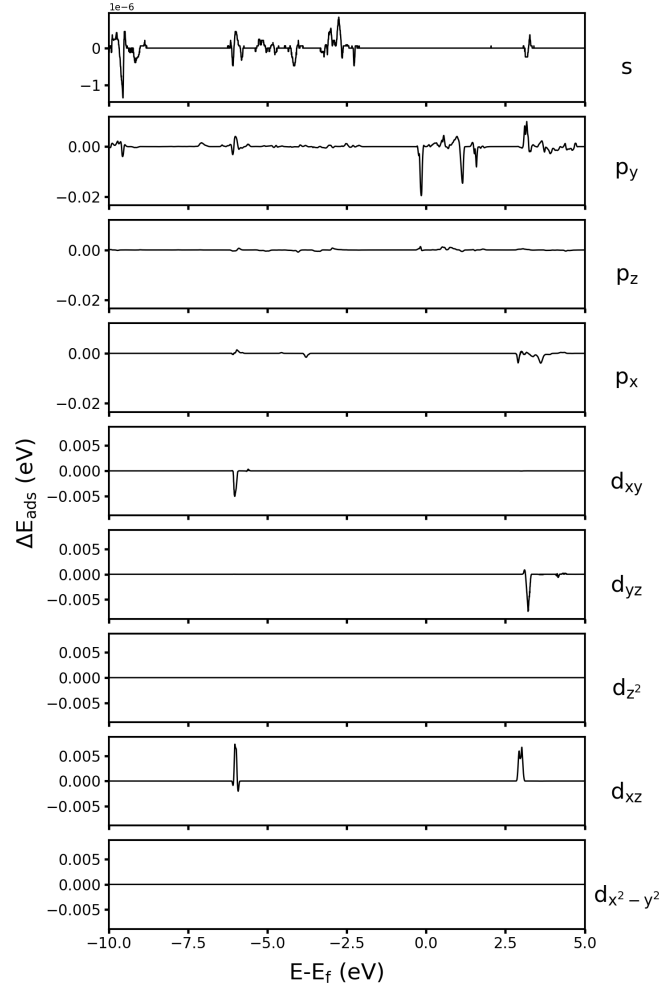
**Figure 27: Illustration of the eDOS occlusion experiment.** This figure demonstrates the step-by-step process of the occlusion experiment. The input eDOS array first undergoes zero-padding, and then a masker of width “width” is applied, resulting in an occlusion array that preserves the shape of the input array.



**Figure 28: Occlusion Experiment with a masker width of “1”.** This figure shows an occlusion experiment on the Ge atom in the initial state of CO adsorption on Ge@g-C<sub>3</sub>N<sub>4</sub>.

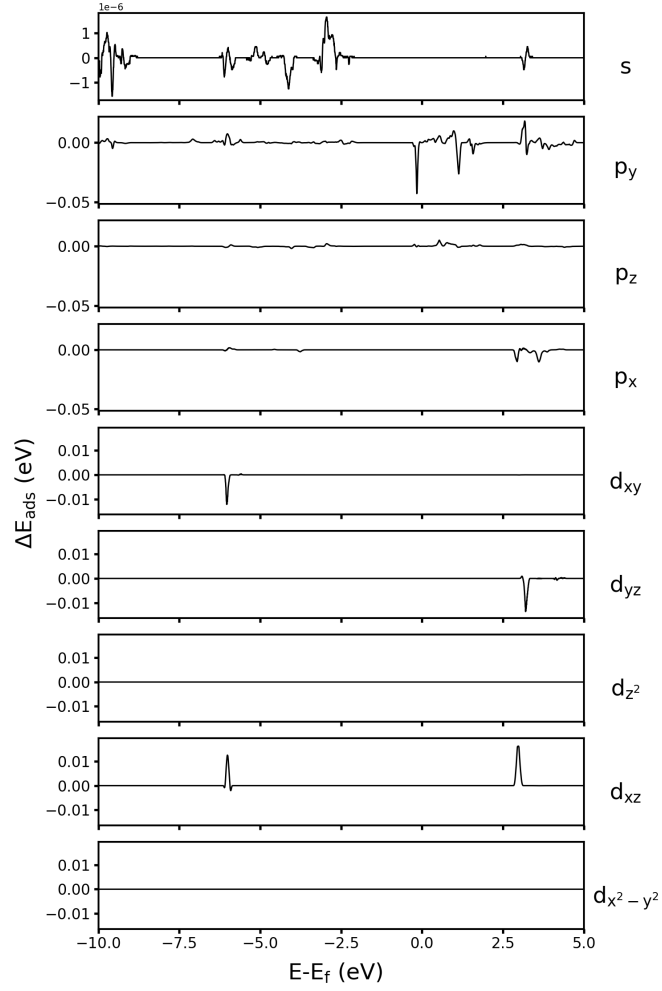


**Figure 29: Occlusion Experiment with a masker width of “3”.** This figure shows an occlusion experiment on the Ge atom in the initial state of CO adsorption on Ge@g-C<sub>3</sub>N<sub>4</sub>.

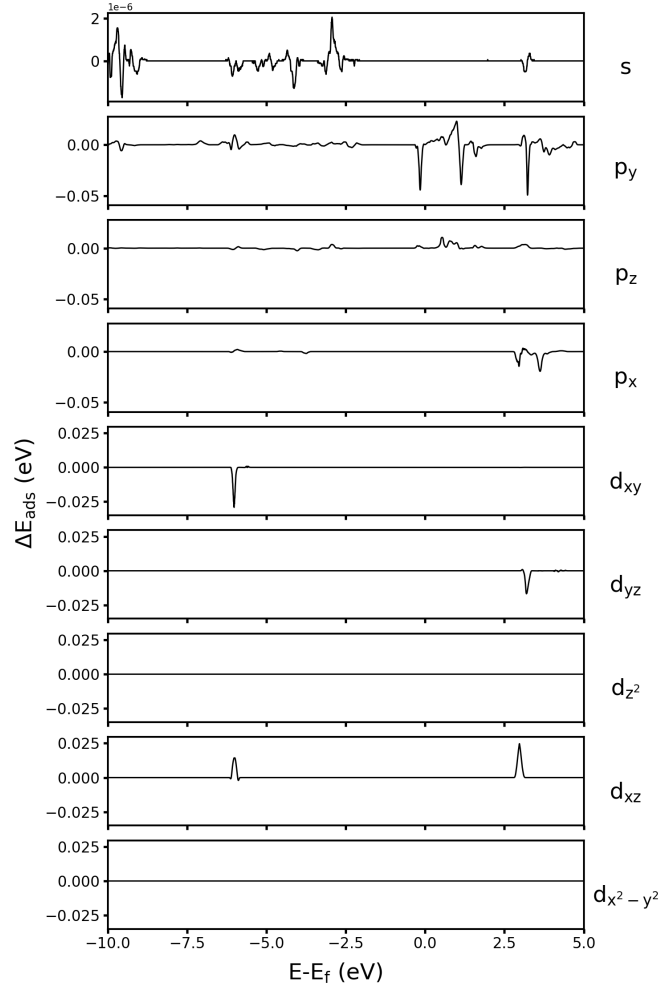


**Figure 30: Occlusion Experiment with a masker width of “5”.** This figure shows an occlusion experiment on the Ge atom in the initial state of CO adsorption on Ge@g-C<sub>3</sub>N<sub>4</sub>.

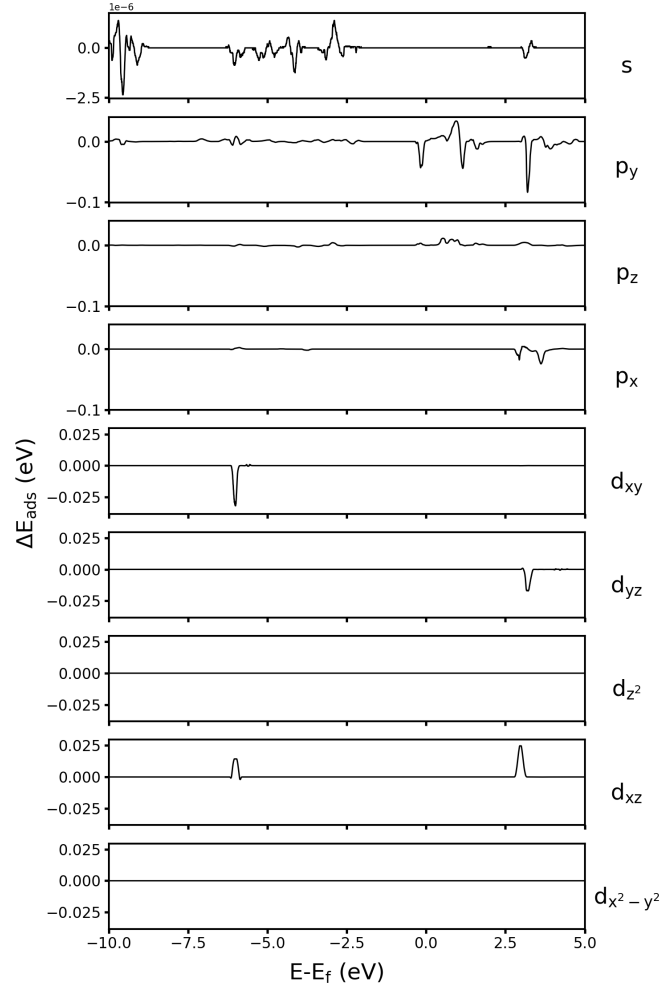




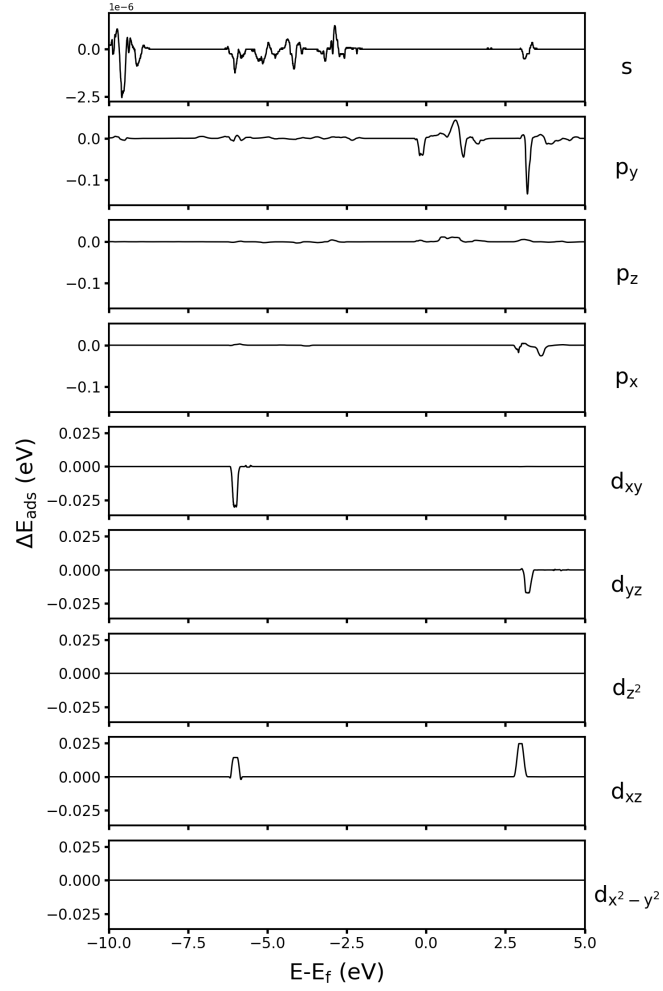
**Figure 31: Occlusion Experiment with a masker width of “11”.** This figure shows an occlusion experiment on the Ge atom in the initial state of CO adsorption on Ge@g-C<sub>3</sub>N<sub>4</sub>.



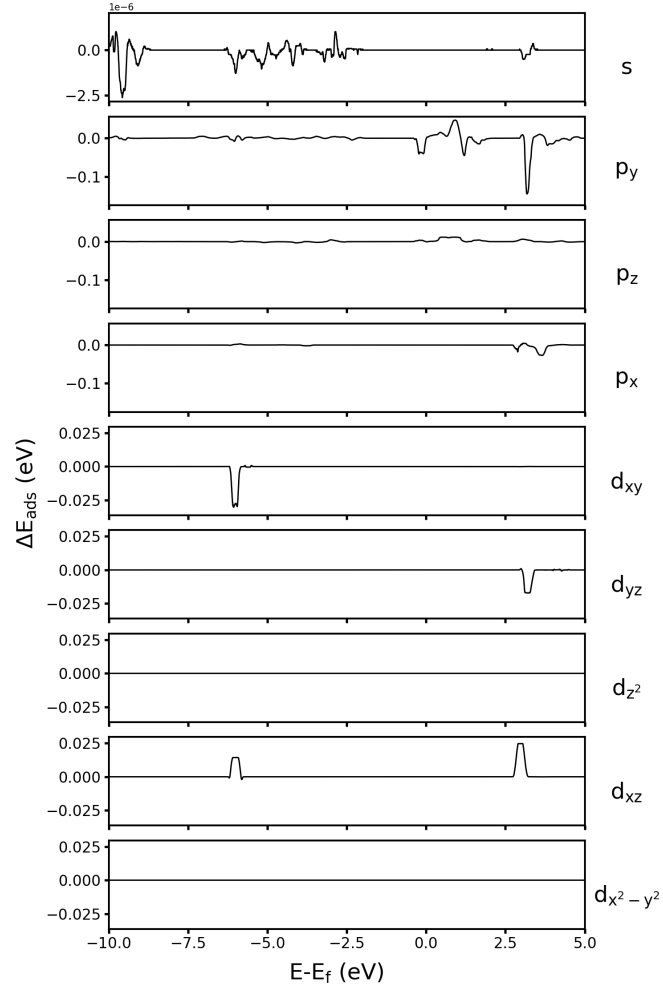
**Figure 32: Occlusion Experiment with a masker width of “21”.** This figure shows an occlusion experiment on the Ge atom in the initial state of CO adsorption on Ge@g-C<sub>3</sub>N<sub>4</sub>.



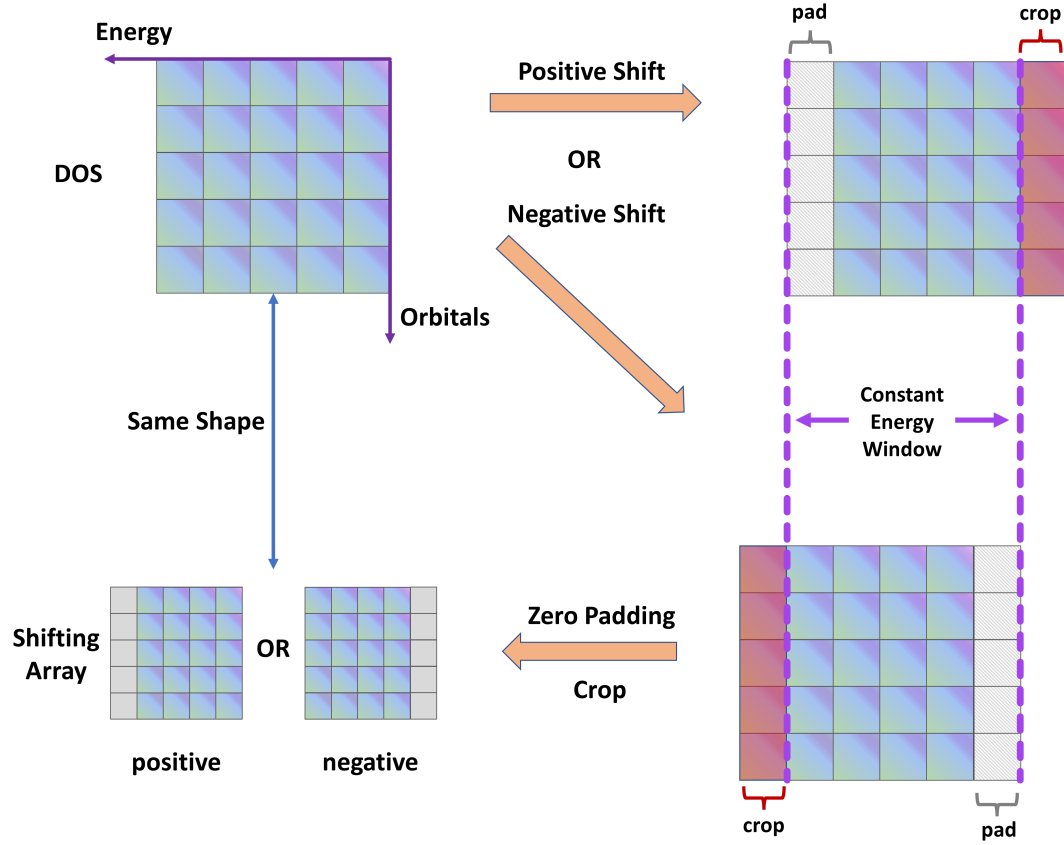
**Figure 33: Occlusion Experiment with a masker width of “31”.** This figure shows an occlusion experiment on the Ge atom in the initial state of CO adsorption on Ge@g-C<sub>3</sub>N<sub>4</sub>.



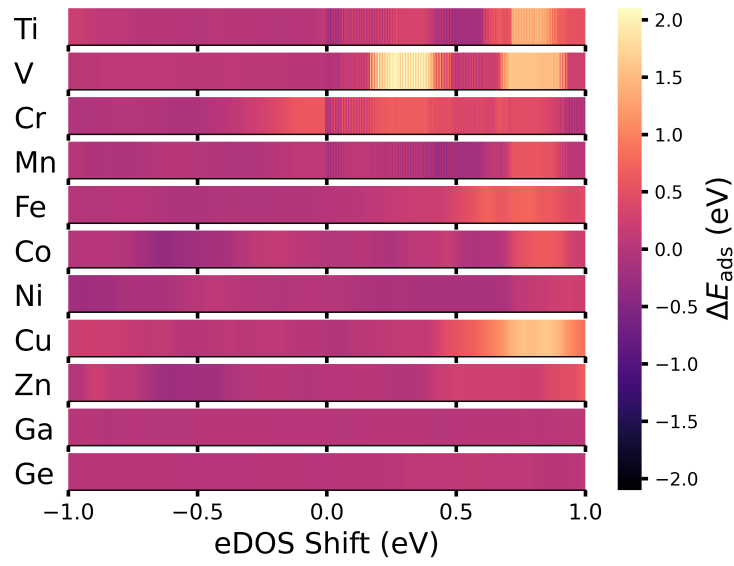
**Figure 34: Occlusion Experiment with a masker width of “41”.** This figure shows an occlusion experiment on the Ge atom in the initial state of CO adsorption on Ge@g-C<sub>3</sub>N<sub>4</sub>.



**Figure 35: Occlusion Experiment with a masker width of “51”.** This figure shows an occlusion experiment on the Ge atom in the initial state of CO adsorption on Ge@g-C<sub>3</sub>N<sub>4</sub>.



**Figure 36: Illustration of the eDOS shifting experiment.** This figure illustrates the sequential stages of the shifting experiment. Initially, the input eDOS array is shifted along the Energy axis. Subsequently, it undergoes zero-padding and cropping, ensuring a consistent shape based on the shifting direction, to match the input eDOS array. The resulting shifted array is then processed by the CNN model to assess the disturbance in adsorption energy caused by the shifting operation.



**Figure 37: Shifting experiment on p orbital.** Effect of orbital shifting on CO adsorption energy, predicted by the CNN model, for single metal atom catalysts supported on g-C<sub>3</sub>N<sub>4</sub>. The perturbations caused by shifting of entire p orbital are presented. The shifting step size corresponds to the energy resolution of the eDOS, set at 0.005 eV in this study. A positive eDOS shift indicates a shift towards higher energy levels, and vice versa.

### 3 Other additional information

#### 3.1 Additional information on machine learning environment

The hyperparameter optimization for the proposed CNN model was performed using NVIDIA® V100 Tensor Core GPUs, courtesy of the Australian National Computational Infrastructure’s (NCI) HPC Gadi system. The HyperBand algorithm [8], as implemented in KerasTuner [9], facilitated the tuning process.

**Table 19:** Core packages in the deep learning environment.

| Name                | Version | Build          | Channel     |
|---------------------|---------|----------------|-------------|
| conda               | 4.13.0  | py39h06a4308_0 | anaconda    |
| cuda-nvcc           | 11.8.89 | 0              | nvidia      |
| cuda-toolkit        | 11.8.0  | 0              | nvidia      |
| cudatoolkit         | 11.6.0  | habf752d_9     | nvidia      |
| ipykernel           | 6.9.1   | py39h06a4308_0 | anaconda    |
| ipython             | 7.34.0  | pypi_0         | pypi        |
| jupyter_client      | 7.2.2   | py39h06a4308_0 | anaconda    |
| jupyterlab          | 3.4.4   | py39h06a4308_0 | anaconda    |
| keras               | 2.9.0   | pypi_0         | pypi        |
| keras-preprocessing | 1.1.2   | pypi_0         | pypi        |
| keras-tuner         | 1.3.5   |                | pypi        |
| matplotlib          | 3.5.1   | py39h06a4308_1 | anaconda    |
| numpy               | 1.22.4  | pypi_0         | pypi        |
| pandas              | 1.3.4   | py39h8c16a72_0 | anaconda    |
| python              | 3.9.12  | h12debd9_1     | anaconda    |
| scikit-learn        | 1.1.1   | py39h6a678d5_0 | anaconda    |
| scipy               | 1.8.1   | py39hddc5342_3 | conda-forge |
| tensorboard         | 2.9.1   | pypi_0         | pypi        |
| tensorflow          | 2.9.3   | pypi_0         | pypi        |
| tensorflow-gpu      | 2.9.3   | pypi_0         | pypi        |
| xlrd                | 2.0.1   | pyhd3eb1b0_0   | anaconda    |
| yaml                | 0.2.5   | h7b6447c_0     | anaconda    |

*Note:* The complete list of packages is provided along with the source code.



### 3.2 Additional information on data analysis and visualization environment

Data analysis and visualization tasks were executed on an Apple MacBook Air featuring an octa-core M2 chip. For accuracy, the TensorFlow-Metal plugin for Mac-based GPU acceleration was deliberately excluded, as it could potentially yield erroneous CNN model predictions.

The illustrations of CNN architecture displayed in Figure 1 of the main text, as well as Figure 27 and Figure 36, were obtained from the GitHub repository `dair-ai/ml-visuals` [10]. Blender was used to create and render the visual representation of the catalyst analysis pipeline shown in Figure 1 of the main text.

**Table 20:** Core packages in data analysis and visualization environment.

| Name             | Version | Build      | Channel |
|------------------|---------|------------|---------|
| bokeh            | 3.2.2   | pypi_0     | pypi    |
| colorcet         | 3.0.1   | pypi_0     | pypi    |
| heatmapz         | 0.0.4   | pypi_0     | pypi    |
| keras            | 2.13.1  | pypi_0     | pypi    |
| keras-tuner      | 1.3.5   | pypi_0     | pypi    |
| matplotlib       | 3.7.2   | pypi_0     | pypi    |
| numpy            | 1.24.3  | pypi_0     | pypi    |
| pandas           | 2.0.3   | pypi_0     | pypi    |
| python           | 3.11.4  | hb885b13_0 |         |
| pyyaml           | 6.0.1   | pypi_0     | pypi    |
| scikit-learn     | 1.3.0   | pypi_0     | pypi    |
| scipy            | 1.11.1  | pypi_0     | pypi    |
| seaborn          | 0.12.2  | pypi_0     | pypi    |
| tensorboard      | 2.13.0  | pypi_0     | pypi    |
| tensorflow       | 2.13.0  | pypi_0     | pypi    |
| tensorflow-macos | 2.13.0  | pypi_0     | pypi    |

*Note:* The complete list of packages is provided along with the source code.

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